Gaia DR3 RR Lyrae Stars to an All-Sky Reddening Map

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We present a comprehensive analysis using Gaia DR3 RR Lyrae variable stars as photometric standard candles to construct an empirical all-sky dust reddening map. Starting from a sample of 100 high-quality Gaia DR3 RR Lyrae stars, we recover pulsation periods via Lomb-Scargle periodograms, finding 89 of 100 agree with Gaia catalog fundamental pulsation frequency values to better than 1%. Fourier series fits (order K selected by cross-validation) reduce the mean magnitude RMS from 0.0282 to 0.0139 mag compared to naïve epoch-based flux-space averaging. A calibration sample of 1031 nearby ($d < 4$ kpc), high-latitude ($|b| > 30°$) stars is assembled from Gaia DR3 and progressively cleaned through astrometric quality cuts (C1/C2; 1031→973) and a mixture contamination model (973→874 stars). Bayesian period-luminosity (PL) relations are fit via three MCMC approaches: Metropolis-Hastings, NUTS with a custom potential, and native PyMC yielded consistent results. For the G-band, we find slopes $a = -2.23 \pm 0.13$ (RRab) and $a = -2.21 \pm 0.18$ (RRc). WISE $W2$ PL fits reveal steeper slopes ($a = -2.99 \pm 0.11$ for RRab), confirming reduced extinction sensitivity in the infrared. Period-color relations in $G_{\mathrm{BP}} - G_{\mathrm{RP}}$ are used to construct color excess maps. After quality filtering, 131,213 stars provide an all-sky reddening map with Pearson $R^2 = 0.457$ correlation with the SFD dust map, validating RR Lyrae stars as effective three-dimensional dust probes.

1. Introduction

Mapping the distribution of interstellar dust is quintessential in modern astronomy. Dust absorbs and scatters starlight, biasing photometric distance estimates and distorting our view of the Galaxy (Schlegel et al. 1998). Traditional two-dimensional dust maps (Schlegel et al. 1998; Schlafly & Finkbeiner 2011) integrate extinction along the entire line of sight, providing no depth information. Three-dimensional dust mapping requires large samples of standard candles with well-characterized colors at known distances.

RR Lyrae stars are ideal for this purpose. As horizontal branch stars undergoing radial pulsations within the classical instability strip, they are among the oldest ($\gtrsim$ 10 Gyr) and most ubiquitous tracers of old stellar populations in the Galaxy (Catelan & Smith 2015). Their well-defined period-luminosity-metallicity relations make them reliable standard candles, and their pulsation period is directly linked to intrinsic color, enabling straightforward dust extinction estimates without spectroscopy.

The advent of the Gaia satellite (Gaia Collaboration et al. 2016) has brought in a treasure trove of high-quality astrometric and photometric data. Gaia Data Release 3 (Vallenari et al. 2023) provides homogeneous light curves, catalog periods, and sub-milliarcsecond parallaxes for hundreds of thousands of RR Lyrae stars across the entire sky (Clementini et al. 2023). This dataset allows for statistical studies of RR Lyrae populations and, as we demonstrate here, their use as dust probes along individual lines-of-sight.

In this work, we develop a complete pipeline from raw Gaia light curve data to an all-sky reddening map. We (1) recover pulsation periods from time-series photometry using the Lomb-Scargle periodogram and Fourier series modeling; (2) construct a clean calibration sample of nearby, high-galactic-latitude low-extinction RR Lyrae stars; (3) fit period-luminosity and period-color relations using three Bayesian Markov Chain Monte Carlo (MCMC) approaches; (4) extend to WISE infrared photometry to characterize wavelength dependence; and (5) apply the period-color calibration to 131,213 RR Lyrae stars across the sky to construct an empirical reddening map, validated against the Schlegel-Finkbeiner-Davis (SFD) map (Schlegel et al. 1998). Figure 1 provides an overview of the full pipeline.

2. Theoretical Background

2.1. RR Lyrae

RR Lyrae stars are low-mass ($\sim 0.6\,M_\odot$), metal-poor, evolved stars on the horizontal branch (HB). After exhausting core hydrogen on the main sequence, these stars settled on the HB with helium-burning cores. Those HB stars that fall within the classical instability strip in the Hertzsprung-Russell diagram undergo radial pulsations driven by the κ-mechanism (Catelan & Smith 2015).

RR Lyrae stars are classified by their pulsation mode:

- **RRab**: fundamental-mode pulsators with periods $P \approx 0.3$–1.0 days, characterized by large amplitudes ($\Delta G \sim 0.5$–1.5 mag) and asymmetric, sawtooth-shaped light curves with a rapid rise and slower decline.
- **RRc**: first-overtone pulsators with shorter periods ($P \approx 0.2$–0.4 days), smaller amplitudes

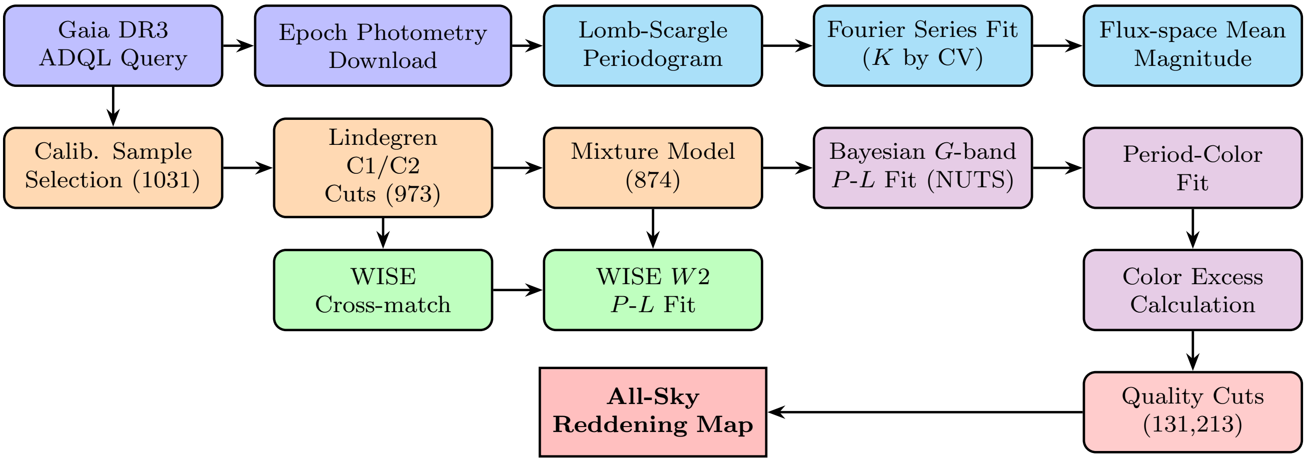

Fig. 1.— Complete analysis pipeline from Gaia DR3 data acquisition to all-sky reddening map. Node color indicates pipeline stage: Data Access (Gaia DR3 query and epoch photometry download); Light Curves (Lomb-Scargle periodogram, Fourier series fitting, flux-space mean magnitude); Calibration (sample selection and astrometric quality cuts); Optical Fits (Bayesian G-band P-L fit, period-color relation, color excess); Infrared Branch (WISE cross-match and $W2$ P-L fit); Dust Map (quality filtering and all-sky reddening map).

($\Delta G \sim 0.2$–0.5 mag), and more sinusoidal light curves.

- **RRd**: double-mode pulsators exciting both fundamental and first-overtone modes simultaneously; much rarer.

The Blazhko effect refers to the observed quasi-periodic modulations of the light curve amplitude and phase on timescales of tens to hundreds of days (Netzel et al. 2018). Stars exhibiting the Blazhko effect show residuals from simple periodic models, complicating period determination.

2.2. Period-Luminosity Relations

The Leavitt law (Leavitt & Pickering 1912) for classical Cepheids demonstrated that pulsation period correlates with intrinsic luminosity. RR Lyrae stars follow an analogous but weaker period-luminosity relation (Catelan & Smith 2015):

$$M_\lambda = a_\lambda \log P + b_\lambda + Z_\lambda [\mathrm{Fe/H}], \tag{1}$$

where M_λ is absolute magnitude in band λ, P is the pulsation period, and [Fe/H] is metallicity. The metallicity dependence Z_λ is significant in optical bands (e.g. Gaia band G) but diminishes in the infrared (e.g. WISE band $W2$).

To note, PL relations have wavelength dependence whereby slopes are steeper and scatter is smaller in the infrared. This is because dust extinction decreases with wavelength ($A_\lambda \propto \lambda^{-\alpha}$, $\alpha \sim 1.7$), so infrared PL relations are less sensitive to residual extinction in the calibration sample. WISE band $W2$ (4.6 μm) offers the additional advantage that the R_{W2} extinction coefficient is nearly negligible (Klein et al. 2014).

2.3. Lomb-Scargle Periodogram

Gaia light curves are irregularly sampled due to the satellite's observation strategy. The standard Fourier transform assumes uniform sampling, so we use the Lomb-Scargle (LS) periodogram (Lomb 1976; Scargle 1982), which generalizes frequency analysis to unevenly sampled data.

For a time series $\{t_i, y_i, \sigma_i\}$, the LS periodogram is defined as (VanderPlas 2018):

$$P_{\mathrm{LS}}(\omega) = \frac{1}{2\hat{\chi}^2}\left[\frac{\left(\sum_i w_i y_i \cos\omega(t_i-\tau)\right)^2}{\sum_i w_i \cos^2\omega(t_i-\tau)} + \frac{\left(\sum_i w_i y_i \sin\omega(t_i-\tau)\right)^2}{\sum_i w_i \sin^2\omega(t_i-\tau)}\right], \tag{2}$$

where $w_i \propto \sigma_i^{-2}$ are measurement weights, $\hat{\chi}^2$ is the weighted variance of the data, and τ is a phase offset chosen to orthogonalize the sine and cosine terms. The LS power $P_{\mathrm{LS}}(\omega) \in [0,1]$ measures the fraction of variance explained by a sinusoid at angular frequency $\omega = 2\pi/P$. The period estimate is the frequency at maximum power.

False alarm probabilities (FAPs) quantify the significance of peaks against noise fluctuations. For the datasets considered here (Gaia light curves with $\sim$30–100 epochs), the highest LS peak is typically highly significant (FAP $\ll 10^{-10}$).

2.4. Fourier Series Modeling

A single sinusoid often poorly fits RR Lyrae light curves, particularly for RRab stars. Hence, we model the light curve as a truncated Fourier series of order K:

$$G(\phi) = A_0 + \sum_{k=1}^{K}\left[A_k \cos(2\pi k\phi) + B_k \sin(2\pi k\phi)\right], \tag{3}$$

where $\phi = (t - t_0)/P \bmod 1$ is the pulsation phase. This can be written in matrix form as $\mathbf{G} = \mathbf{X}\boldsymbol{\beta}$, where the design matrix $\mathbf{X}$ has rows:

$$\mathbf{x}_i = \big[1,\, \cos(2\pi\phi_i),\, \sin(2\pi\phi_i),\, \ldots, \\ \cos(2\pi K\phi_i),\, \sin(2\pi K\phi_i)\big]. \tag{4}$$

Stacking all N rows gives the full design matrix and coefficient vector:

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_N \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} a_0 \\ a_1 \\ b_1 \\ \vdots \\ a_K \\ b_K \end{pmatrix}. \tag{5}$$

The least-squares solution is $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}\mathbf{G}$, where $\mathbf{W} = \mathrm{diag}(\sigma_i^{-2})$.

The per-epoch weights $\sigma_{G,i}$ combine the Gaia epoch flux signal-to-noise and the G-band zero-point uncertainty σ_{ZP}:

$$\sigma_{G,i} = \sqrt{\left(\frac{2.5}{\ln 10}\,\frac{\sigma_{f,i}}{f_i}\right)^2 + \sigma_{\mathrm{ZP}}^2}. \tag{6}$$

The formal coefficient covariance is:

$$C_{\boldsymbol{\beta}} = \max\big(\chi^2_\nu,\, 1\big)\,(\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}, \tag{7}$$

where $\chi^2_\nu = (N - 2K - 1)^{-1}\sum_i (G_i - \hat{G}_i)^2/\sigma^2_{G,i}$. All downstream Fourier uncertainties (predicted magnitudes, mean magnitude) are propagated from $C_{\boldsymbol{\beta}}$.

The order K is chosen by cross-validation (CV) where we split the light curve into training ($\sim$80%) and validation ($\sim$20%) sets, fit on the training set for each K, and select the K that minimizes the validation reduced χ^2_r:

$$\chi^2_r = \frac{1}{N_{\mathrm{CV}}}\sum_{i\in\mathrm{CV}} \frac{(G_i - \hat{G}_i)^2}{\sigma^2_{G,i}}. \tag{8}$$

The mean magnitude of the star is computed in two ways. The first is the epoch mean $\langle G\rangle_{\mathrm{epoch}}$, obtained by averaging the epoch fluxes and converting back to magnitudes:

$$\langle G\rangle_{\mathrm{epoch}} = -2.5\log_{10}\left(\frac{1}{N}\sum_{i=1}^{N} f_i\right) + ZP_G, \tag{9}$$

where f_i are the Gaia epoch fluxes (`g_transit_flux`) and ZP_G is the G-band zero-point. The second is the Fourier mean $\langle G\rangle_{\mathrm{Fourier}}$, computed in flux space to avoid the non-linearity of the magnitude scale:

$$\langle G\rangle_{\mathrm{Fourier}} = -2.5\log_{10}\left[\frac{1}{2\pi}\int_0^{2\pi} 10^{-0.4\,G(\phi)}\,d\phi\right]. \tag{10}$$

This integral is evaluated numerically from the Fourier fit coefficients, and its uncertainty is propagated from $C_{\boldsymbol{\beta}}$ by linearising $\langle G\rangle_{\mathrm{Fourier}}$ around $\hat{\boldsymbol{\beta}}$ and $\sigma^2_{\langle G\rangle} \approx (\nabla_{\boldsymbol{\beta}}\langle G\rangle)^T C_{\boldsymbol{\beta}}\,(\nabla_{\boldsymbol{\beta}}\langle G\rangle)$, with the gradient evaluated numerically over a phase grid.

2.5. Bayesian MCMC Inference

We fit the PL relation using Bayesian inference. The posterior distribution over parameters $\boldsymbol{\theta} = (a, b, \sigma_{\mathrm{scatter}})$ is:

$$p(\boldsymbol{\theta} \mid \mathbf{y}) \propto p(\mathbf{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}), \tag{11}$$

where $p(\mathbf{y} \mid \boldsymbol{\theta})$ is the likelihood and $p(\boldsymbol{\theta})$ the prior.

The likelihood accounts for both measurement uncertainties σ_i and intrinsic astrophysical scatter $\sigma_{\mathrm{scatter}}$ (due to metallicity spread, depth effects, etc.). Defining $\delta_i \equiv \log P_i - \langle \log P\rangle_{\mathrm{class}}$ as the centered log-period:

$$\log\mathcal{L} = -\frac{1}{2}\sum_i \left[\frac{(M_{G,i} - a\,\delta_i - b)^2}{\sigma_i^2 + \sigma^2_{\mathrm{scatter}} + a^2\sigma^2_{\log P,i}} \right. \\ \left. + \log(2\pi(\sigma_i^2 + \sigma^2_{\mathrm{scatter}} + a^2\sigma^2_{\log P,i}))\right]. \tag{12}$$

Here $\sigma^2_{M,i} = \sigma^2_{G,i} + \sigma^2_{\mu,i}$ combines the apparent-magnitude and distance-modulus uncertainties in quadrature. The apparent-magnitude error follows the same form as Eq. (6), with f_i and $\sigma_{f,i}$ replaced by the catalog mean flux F_i (`phot_g_mean_flux`) and its uncertainty $\sigma_{F,i}$ (`phot_g_mean_flux_error`). The distance-modulus error propagates the Gaia parallax uncertainty $\sigma_{\varpi,i}$:

$$\sigma_{\mu,i} = \frac{5\,\sigma_{\varpi,i}}{\varpi_i \ln 10}. \tag{13}$$

We implement three MCMC samplers:

Metropolis-Hastings (M-H): A random-walk sampler that proposes steps $\boldsymbol{\theta}' = \boldsymbol{\theta} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\mathrm{prop}})$, and accepts with probability $\alpha = \min(1, p(\boldsymbol{\theta}')/p(\boldsymbol{\theta}))$ (Metropolis et al. 1953; Hastings 1970). The per-parameter proposal standard deviations are tuned by scanning a discrete grid and selecting the scale whose acceptance rate is closest to $\sim$0.5.

NUTS (No-U-Turn Sampler): A Hamiltonian Monte Carlo (HMC) variant (Hoffman & Gelman 2014). HMC augments the parameter vector $\boldsymbol{\theta}$ with auxiliary momentum variables $\mathbf{p}$ and simulates Hamiltonian dynamics along the posterior surface whereby the gradient $\nabla \log p(\boldsymbol{\theta} \mid \mathbf{y})$ acts as a force that propels the sampler along high-probability regions, allowing large, nearly uncorrelated steps rather than the local random-walk steps of M-H. NUTS extends HMC by automatically determining how long to run each trajectory where it doubles the path length until the trajectory would start turning back on itself, after which it selects a point from the completed trajectory. Additionally, step sizes are automatically selected during the warm-up phase, removing the manual proposal-width grid search required by M-H. We implement NUTS both with a custom `Potential` term in PyMC and using native PyMC distributions.

2.6. Interstellar Extinction via Period-Color Residuals

Interstellar dust reddens starlight by absorbing and scattering photons. The color excess $E(G_{\mathrm{BP}} - G_{\mathrm{RP}})$

describes the difference between observed and intrinsic color:

$$E(G_{\rm BP} - G_{\rm RP}) = (G_{\rm BP} - G_{\rm RP})_{\rm obs} - (G_{\rm BP} - G_{\rm RP})_{\rm int}(P), \quad (14)$$

where the intrinsic color $(G_{\rm BP} - G_{\rm RP})_{\rm int}(P)$ is predicted from our class-specific period-color calibration (Section 3).

The G-band extinction is then:

$$A_G = R_G \cdot E(G_{\rm BP} - G_{\rm RP}), \quad (15)$$

where $R_G = 2.0$ is the ratio of total-to-selective extinction adopted for the Gaia G band. This value is consistent with the extinction law of Cardelli et al. (1989) as updated by O'Donnell (1994) applied to the Gaia G bandpass, and with the empirical calibration by Wang & Chen (2019) who find $R_G \approx 2.0$ for Gaia DR2 photometry at typical red giant temperatures. Conceptually, since the Gaia DR3 G band is broad, and R_G has mild color dependence, for the range of stellar temperatures relevant to RR Lyrae ($T_{\rm eff} \sim 6000$–7500 K), $R_G \approx 2.0$ is an appropriate approximation.

3. Data Acquisition

3.1. Gaia DR3 Light Curve Sample (100 Stars)

We began by querying the Gaia DR3 `vari_rrlyrae` catalog for 100 high-quality RR Lyrae stars suitable for light curve analysis. The selection prioritized high-epoch-count stars for reliable period recovery:

- `pf IS NOT NULL` (defined fundamental period)
- `num_clean_epochs_g` > 40 (high epoch count for period recovery)
- Ordered by `num_clean_epochs_g DESC`, limiting to 100 stars

The query was submitted via the Gaia TAP service using `astroquery.gaia`. Epoch photometry (time, magnitude, flux, uncertainty) for each star was then downloaded individually from the Gaia epoch photometry tables.

3.2. Calibration Sample (1031 Stars)

For the period-luminosity calibration, we required a sample of nearby, low-extinction RR Lyrae stars with precise Gaia parallaxes. The calibration sample was selected with:

- `parallax_over_error` > 5 (relative parallax error < 20%)
- $|b| > 30°$ (high Galactic latitude, low extinction)
- Distance $d < 4$ kpc (derived from $\varpi > 0.25$ mas)

This returned 1031 stars with sufficient data for Bayesian PL fitting.

3.3. Full All-Sky Catalog (269,772 Stars)

For dust mapping, we downloaded the complete Gaia DR3 RR Lyrae catalog without distance restrictions. This returned 269,772 stars, forming the basis for the all-sky reddening map.

3.4. WISE W2 Cross-Match

Infrared photometry in WISE $W2$ (4.6 μm) was retrieved for calibration sample stars via the Gaia DR3 `allwise_best_neighbour` table. This cross-match links Gaia sources to their nearest AllWISE counterpart within a 1 arcsec radius. Referemce Gaia DR3 stars were required to have a unique best WISE neighbour, finite $W2$ magnitude and uncertainty, photometric quality flag `ph_qual[1]` of `A` or `B`, no contamination flag (`cc_flags[1] = '0'`), and no extended-source flag (`ext_flag` ≤ 1).

4. Light Curve Analysis

Throughout this section we use an RRab sdtar Gaia DR3 source `4659759557323962752` as a running example to illustrate each analysis step.

4.1. Period Recovery via Lomb-Scargle

We applied the Lomb-Scargle periodogram to the G-band epoch photometry of each of the 100 sample stars. The peak frequency $\omega_{\rm max}$ was identified, and the period $P_{\rm LS} = 2\pi/\omega_{\rm max}$ was recorded.

Figure 2 shows raw and phase-folded light curves, and Figure 3 shows the corresponding LS periodogram.

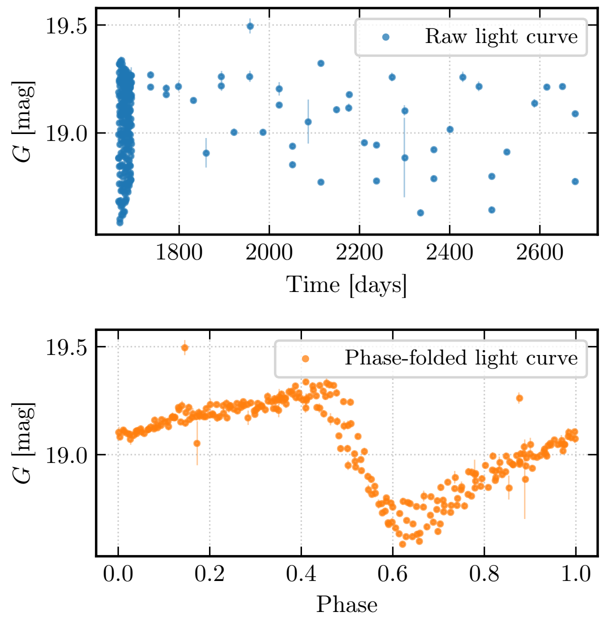

Fig. 2.— Example Gaia DR3 G-band light curves. *Top:* Raw time series with photometric uncertainties. *Bottom:* Phase-folded light curve using the LS-recovered period, showing the characteristic sawtooth morphology of fundamental-mode pulsation.

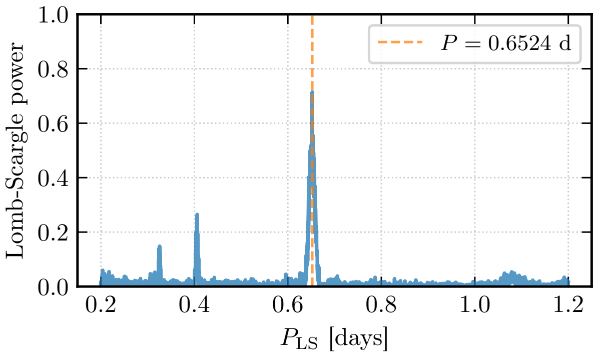

Fig. 3.— Example Lomb-Scargle periodogram. The vertical dashed line marks the peak period $P_{\rm LS}$ adopted for subsequent Fourier fitting.

Figure 4 compares our recovered periods to the Gaia DR3 catalog fundamental periods (`pf`). We find that 89 of 100 stars agree to within 1% ($|P_{\rm LS} - P_{\rm pf}|/P_{\rm pf} < 0.01$). For all 11 outlying stars, inspection of the LS power spectrum reveals a second significant peak at a longer period agreeing to `pf`; the recovered $P_{\rm LS}$ matches the Gaia first-overtone period `p1o` rather than the fundamental `pf`. This identifies them as RRd double-mode pulsators, whose light curves contain power at both P_0 (fundamental) and P_1 (first overtone, $P_1/P_0 \approx 0.745$). The LS periodogram latches onto whichever peak carries the most power, which for these stars is P_1.

In the phase-folded light curves of several stars in the sample, low-amplitude quasi-periodic modulations characteristic of the Blazhko effect are visible as a scatter that is coherent on long timescales but incoherent within a single epoch window. However, the LS power associated with the Blazhko side-lobes (typically $\Delta\nu \sim 1/P_{\rm Blazhko}$ away from the main peak) is orders of magnitude smaller than the main pulsation peak. As such, the Blazhko effect does not displace the highest LS peak and has no effect on our period recovery; its likely only signature is a slight accentuation of the residuals after phase-folding on the recovered period.

4.2. Fourier Series Fitting and Cross-Validation

For each star, we fit Fourier series of orders $K = 1, 3, 5, 7, 9$ to the phase-folded light curve using weighted least squares. The design matrix $\mathbf{X}$ for each order was constructed as described in Section 2. Figure 5 shows the progression of fit quality with increasing K for the example star.

The optimal K was selected via cross-validation (Figure 6). The training reduced χ^2_r monotonically decreases with K, while the validation χ^2_r has a minimum at $K \approx 5$–7 for most stars, beyond which overfitting to noise causes the validation error to increase. We adopted the K minimizing validation χ^2_r for each star.

Figures 7 and 8 further diagnose the underfitting and overfitting regimes. Figure 7 shows normalised residuals $(G - G_{\rm model})/\sigma$ for the low-K and best-K fits on both the training and validation sets. In the low-K panel both distributions are broader than the reference $\mathcal{N}(0, 1)$, indicating genuine light-curve structure left in the residuals. At best K the two distributions converge and narrow toward unity, showing the

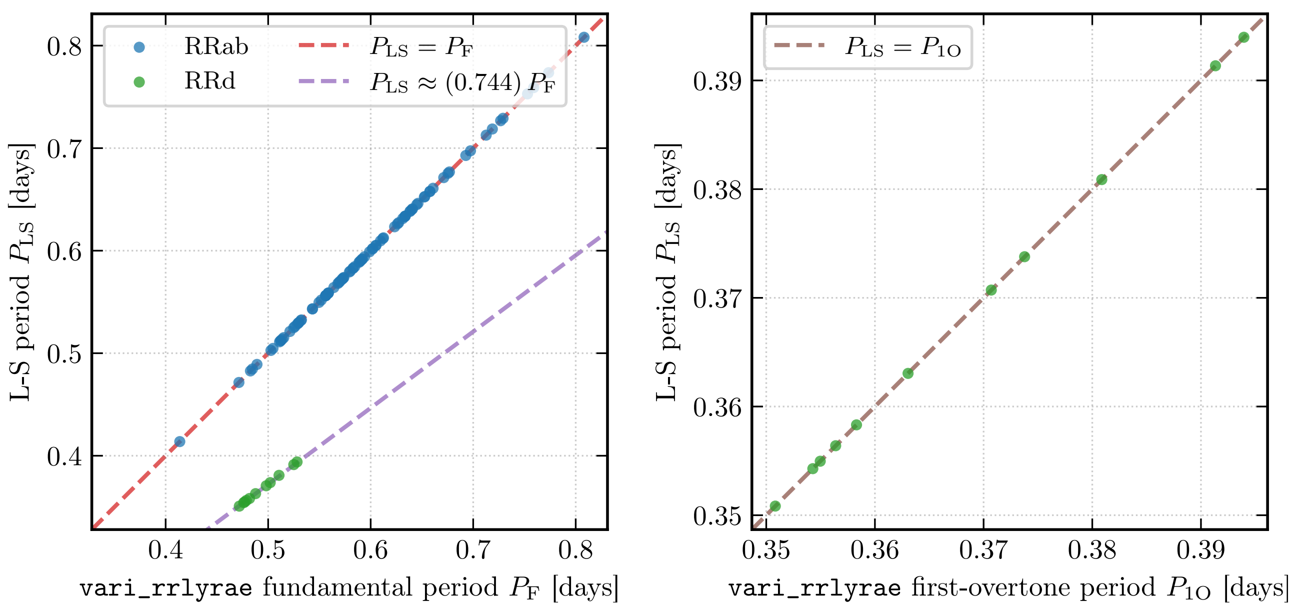

Fig. 4.— Comparison of LS-recovered periods to Gaia DR3 catalog fundamental periods (`pf`) for 100 stars. Points within 1% agreement (89/100) are shown in blue; outlying stars are in red. The diagonal line shows perfect agreement. For all 11 red points, the recovered period matches the Gaia first-overtone period `p1o`, identifying them as RRd double-mode pulsators.

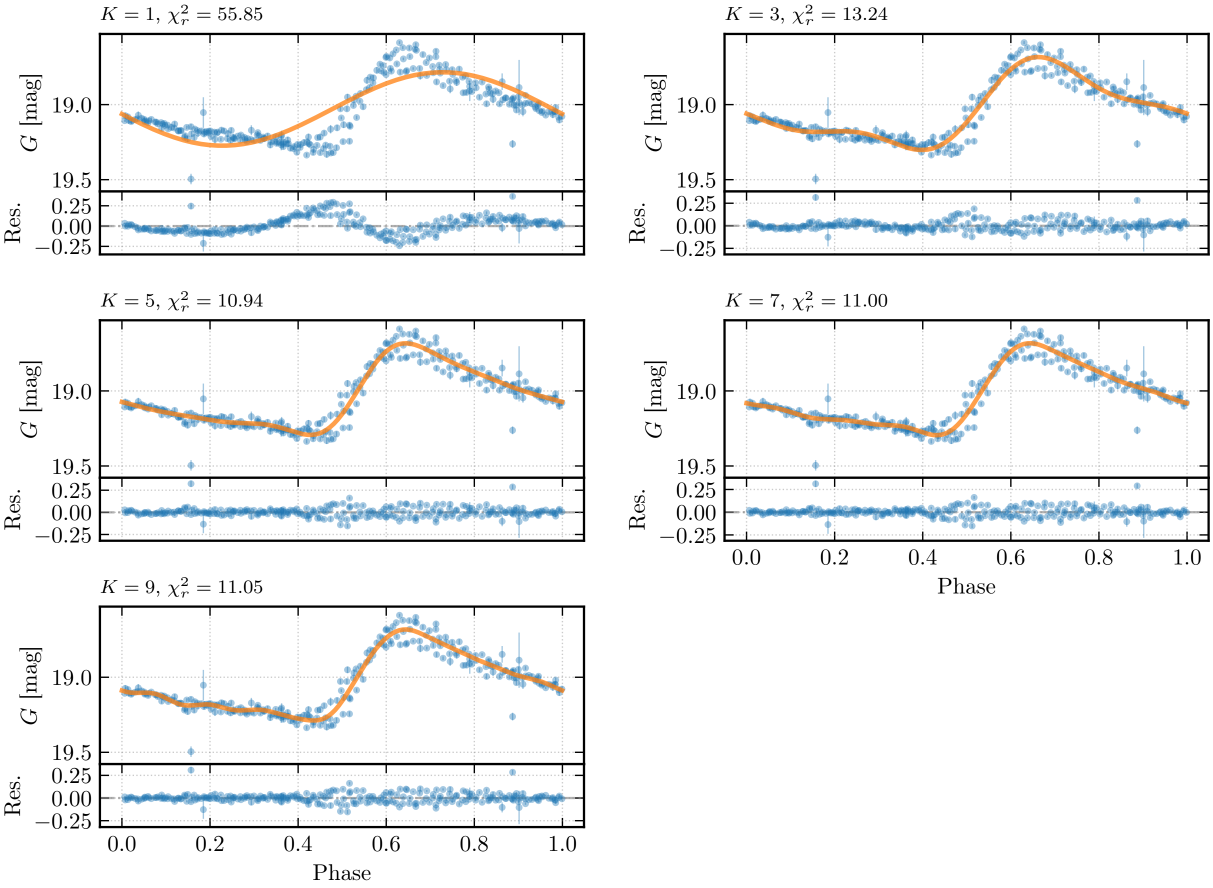

Fig. 5.— Fourier series fits of order $K = 1, 3, 5, 7, 9$ to the phase-folded RRab light curve. Low orders ($K = 1, 3$) fail to capture the rapid rise; $K = 5$ provides a good fit; higher orders begin to overfit noise.

model now captures the dominant astrophysical shape. Figure 8 compares the best-K and high-K phase fits. Here, the high-K model interpolates training-set noise and departs from the validation points.

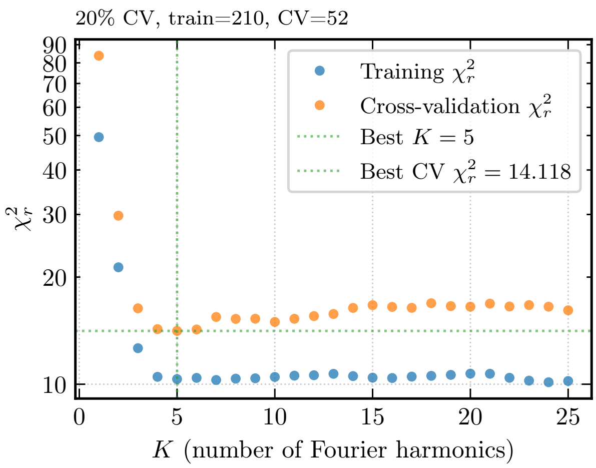

Fig. 6.— Cross-validation curve for the example RRab star where training χ_r^2 (blue) and validation χ_r^2 (orange) as a function of Fourier order K. The optimal $K = 5$ minimizes validation error before overfitting sets in.

4.3. Flux-Space Mean Magnitudes

A Fourier model is a more accurate mean magnitude estimator; averaging magnitudes directly is biased because the relationship between flux and magnitude is nonlinear ($f \propto 10^{-0.4m}$). We instead computed the mean flux from the Fourier model by numerical integration in flux space and converted back to magnitude (see Section 2).

Figure 9 compares both estimators against the Gaia DR3 `int_average_g` (the catalog intensity-averaged magnitude). The left panel shows the flux-space epoch mean, $\langle G \rangle_{\rm epoch}$, against the catalog value; the right panel shows $\langle G \rangle_{\rm Fourier}$. Residual panels below each scatter plot show the signed offset from the catalog. The Fourier-based mean is more tightly clustered around the one-to-one line, with RMS residual 0.0139 mag vs. 0.0282 mag for the epoch mean. The improvement arises because the Fourier template reconstructs the full pulsation cycle, correcting for uneven phase sampling across Gaia visits that inflates the epoch-average scatter.

4.4. Light Curve Extrapolation

The Fourier model, once fit, can predict the star's brightness at arbitrary future times. Figure 10 shows a 10-day extrapolation beyond the last Gaia epoch for

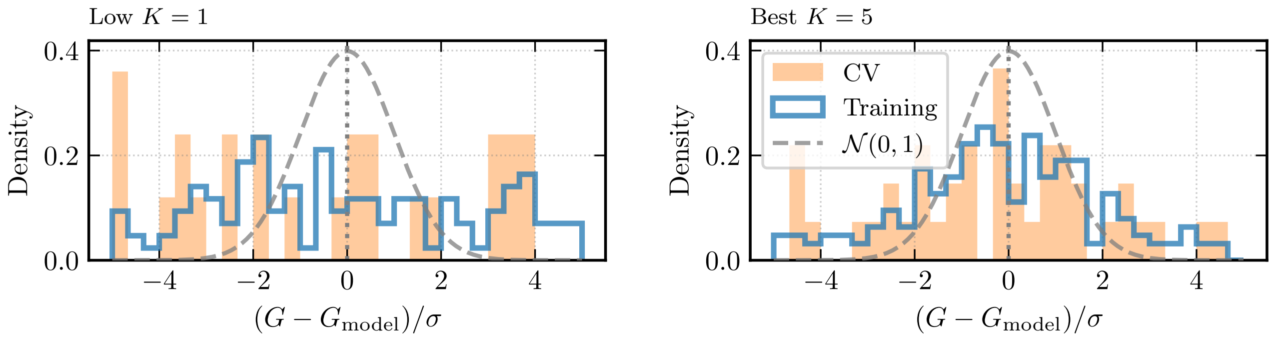

Fig. 7.— Normalised residual histograms $(G - G_{\text{model}})/\sigma$ for the training (blue outline) and validation (orange fill) sets. *Left:* In the low-K underfitting regime, both distributions are broader than $\mathcal{N}(0, 1)$ (dashed), indicating unmodelled light-curve structure. *Right:* In the best-K regime, training and validation residuals converge and approach the reference Gaussian, showing adequate model complexity.

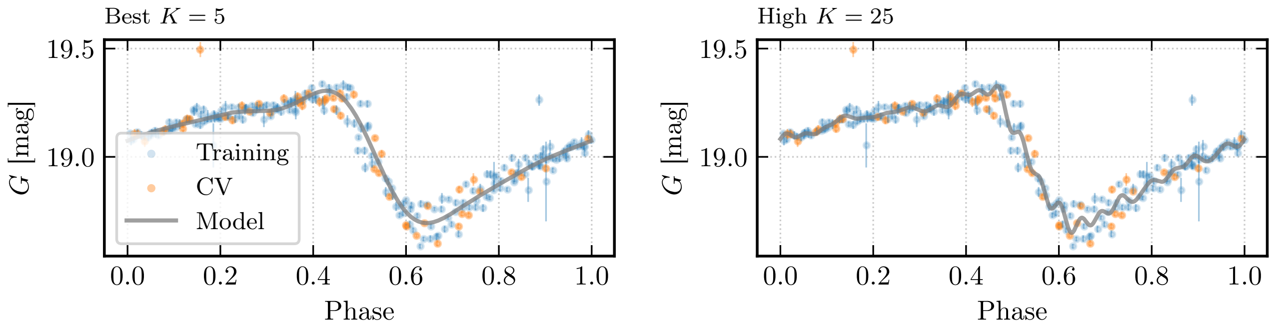

Fig. 8.— Phase-folded light curves for the best-K (left) and high-K (right) Fourier fits. Training epochs are shown in blue and validation epochs in orange. The high-K model overfits training noise and diverges from the validation points.

the running-example star. The shaded band represents the $\pm 1\sigma$ predictive uncertainty propagated from the full Fourier coefficient covariance matrix $\mathbf{C}_\beta$; the vertical dashed line marks the last observed epoch.

4.5. RRab vs. RRc Morphology

Figure 11 illustrates the morphological differences between RRab and RRc light curves. RRab stars (left panels) show the characteristic sawtooth shape consisting of a rapid rise ($\sim$20% of the period) followed by a slower decline, with peak-to-peak amplitudes of 0.5–1.5 mag in G. RRc stars (right panels) are more sinusoidal with smaller amplitudes (0.2–0.5 mag) and shorter periods. These differences reflect the different pulsation modes and instability strip locations (Catelan & Smith 2015).

5. Calibration Sample Construction

5.1. Initial Selection

The initial calibration sample of 1031 RR Lyrae stars (Section 3) spans the northern and southern sky at high Galactic latitudes. Figure 12 shows the sky distribution in Aitoff projection; all stars lie outside the $|b| < 30°$ exclusion zone by construction, confirming that the ADQL query has removed Galactic disk stars. The figure also shows the spatial pattern of stars removed versus retained after the C1/C2 cuts.

5.2. Lindegren Astrometric Quality Cuts

Raw parallaxes from Gaia DR3 can be affected by astrometric systematics, particularly for sources with poor five-parameter astrometric solutions. We applied two quality cuts following Lindegren et al. (2021):

C1 (astrometric unit-weight error): We compute the astrometric unit-weight error $u = \sqrt{\chi^2_{\text{al}}/(n_{\text{obs}} - 5)}$ from the along-scan χ^2 and reject stars with $u > 1.2 \max(1, e^{-0.2(G-19.5)})$. This magnitude-dependent threshold selects stars with well-behaved five-parameter astrometric solutions.

C2 (BP/RP photometric excess): Stars whose BP+RP excess factor E falls outside the two-sided, colour-dependent envelope $1 + 0.015(G_{\text{BP}} - G_{\text{RP}})^2 < E < 1.3 + 0.06(G_{\text{BP}} - G_{\text{RP}})^2$ are flagged as likely blended or contaminated and removed.

After both cuts, the sample reduced from 1031 to 973 stars, a $\sim$5.6% rejection rate. Of the 1031 stars, 998 pass C1 alone and 973 pass C2 alone; only the 973

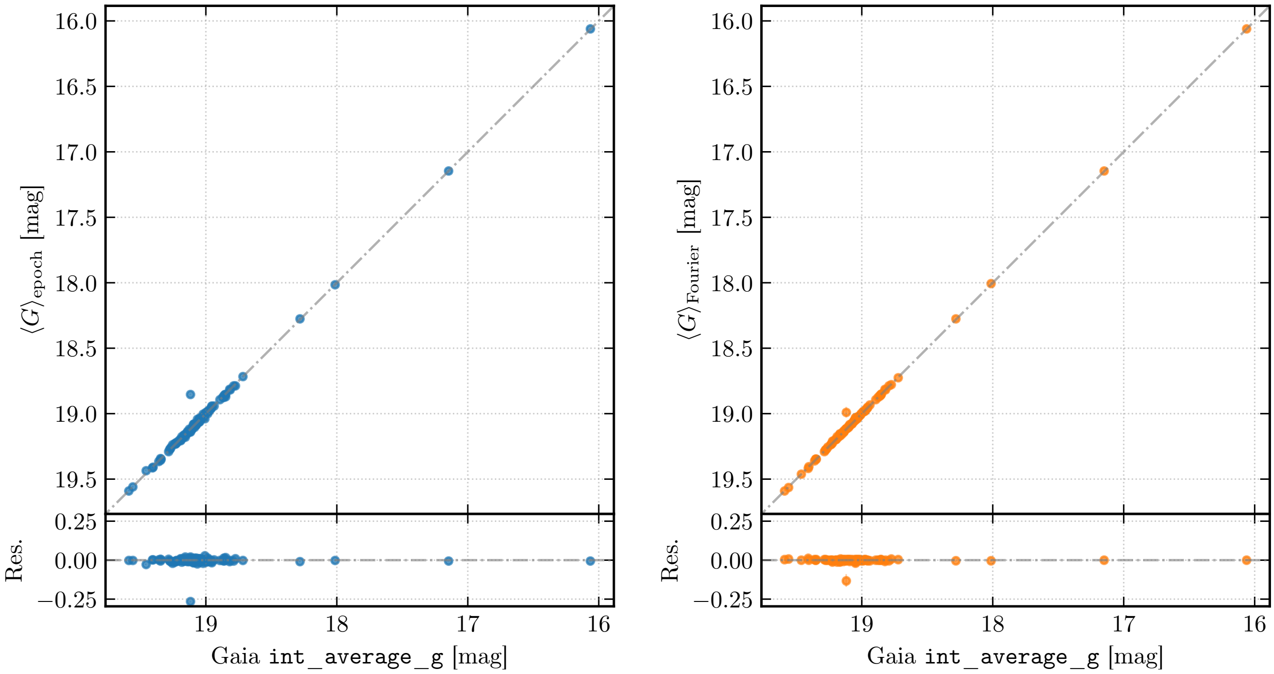

Fig. 9.— Comparison of mean G-band magnitude estimators against the Gaia DR3 `int_average_g` catalog value. *Left:* Flux-space epoch mean $\langle G \rangle_{\rm epoch}$ vs. catalog; *Right:* Fourier mean $\langle G \rangle_{\rm Fourier}$ vs. catalog. Residual panels (bottom) show the signed offset; the dashed line marks zero. The Fourier mean achieves RMS = 0.0139 mag vs. 0.0282 mag for the epoch mean.

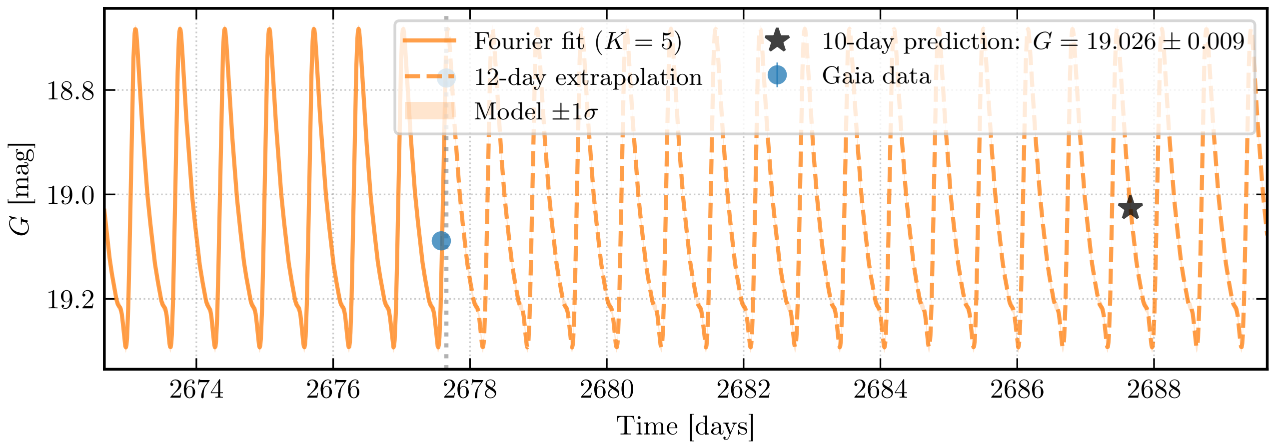

Fig. 10.— Light curve extrapolation for Gaia DR3 source `4659759557323962752`. The final five days of Gaia epoch photometry (grey points with error bars) anchor the best-fit Fourier model (orange curve). The shaded band shows the $\pm1\sigma$ uncertainty from the coefficient covariance $\mathbf{C}_\beta$. The vertical dashed line marks the last observed epoch; the model is extrapolated 10 days into the future.

passing both cuts are retained. Table 1 summarizes the sample evolution through each cleaning stage.

An interesting feature of the removals is their spatial non-uniformity (Figure 12). The most conspicuous concentrations of rejected stars coincide with the directions of the Large and Small Magellanic Clouds (LMC and SMC), where crowding and source blending degrade Gaia's astrometric and photometric quality diagnostics. Because the sample already requires $\varpi > 0.25$ mas, genuine Magellanic Cloud members at $\sim$50–60 kpc are excluded by the parallax selection; the rejected clumps therefore represent problematic Gaia solutions projected against the Cloud directions, not actual Cloud members. This makes their removal by C1/C2 physically sensible. Quantitatively, among the 58 stars removed by C1/C2, those within 10° of the LMC center have a rejection rate of $15/30 = 50.0\%$, while those within 6° of the SMC center show a rejection rate of $7/16 = 43.8\%$. By contrast, stars elsewhere on the sky are removed at only $36/985 = 3.7\%$,

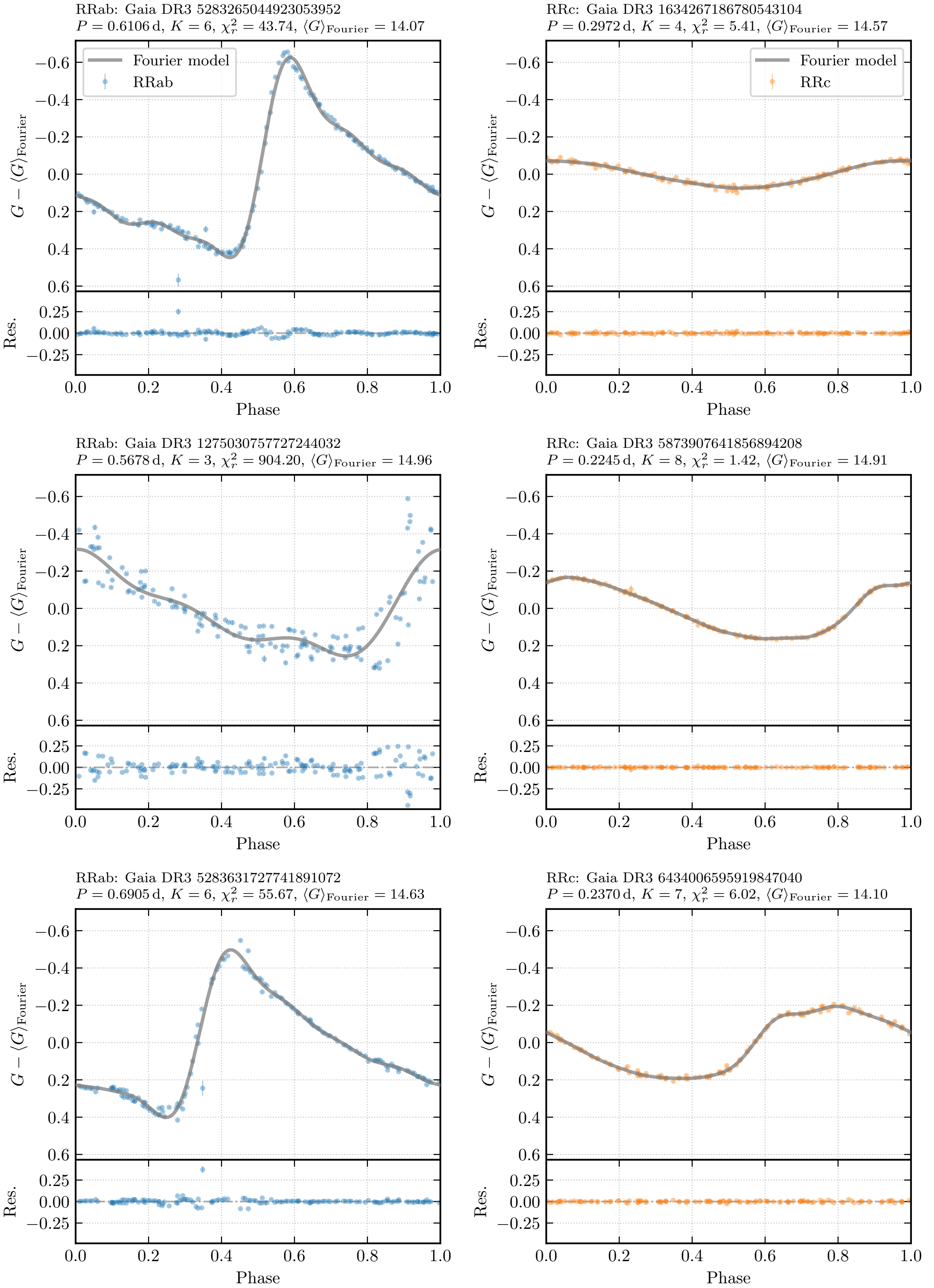

Fig. 11.— Representative phase-folded G-band light curves for three RRab (top) and three RRc (bottom) stars. RRab stars show large-amplitude, asymmetric sawtooth profiles typical of fundamental-mode pulsation; RRc stars are more sinusoidal with smaller amplitudes, consistent with first-overtone pulsation.

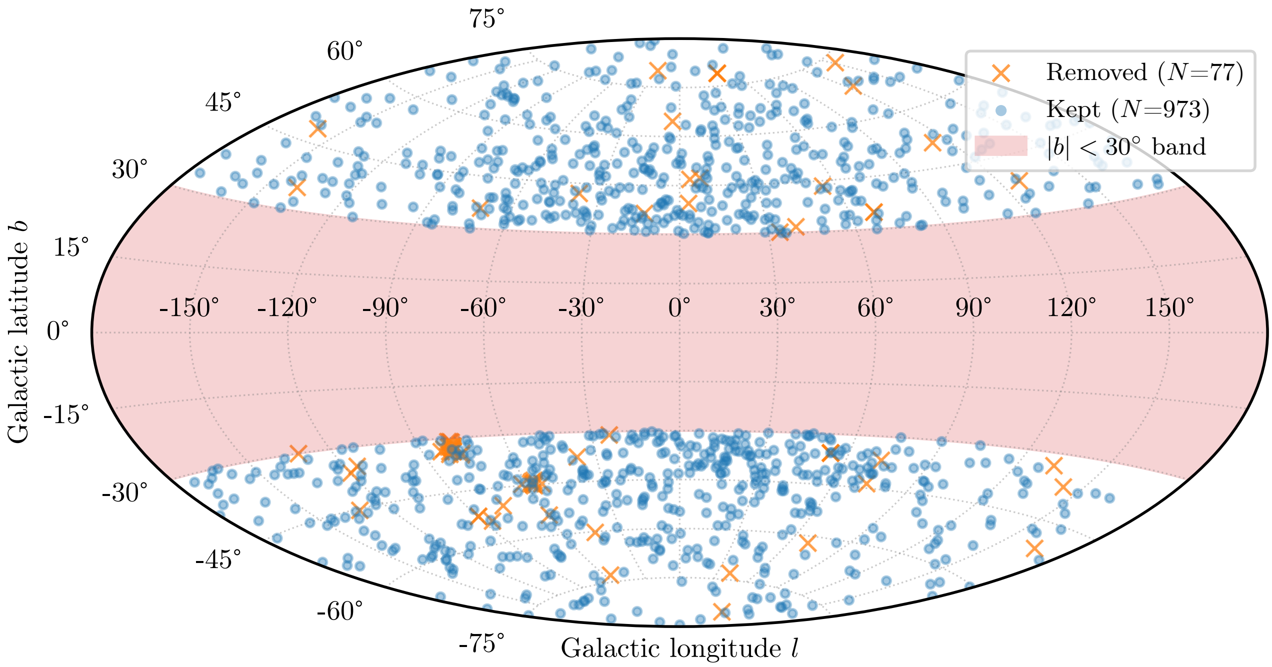

Fig. 12.— Stars removed (orange) versus retained (blue) after the Lindegren C1/C2 quality cuts. Rejected stars cluster toward the LMC ($l \approx 280°$, $b \approx -33°$) and SMC ($l \approx 303°$, $b \approx -44°$) directions, where crowding degrades Gaia astrometric and photometric solutions. These are not Magellanic Cloud members (since the $\varpi > 0.25$ mas cut already excludes stars beyond 4 kpc) but rather problematic catalog solutions in crowded fields.

roughly equal to the basal rejection rate for genuine astrometric outliers. The elevated rejection fractions toward the Magellanic Clouds provide strong empirical evidence that crowding-induced catalog artifacts are concentrated in those directions.

5.3. Mixture Contamination Model

Even after astrometric quality cuts, some outliers remain in the PL plane. These may be genuinely misclassified stars, unresolved binaries affecting the parallax, or stars with anomalous extinction. We modeled the data as a mixture of two populations (Hogg et al. 2010):

- **Inliers** ($f_{\rm in}$): true RR Lyrae stars following the PL relation, with residuals $\epsilon_i \sim \mathcal{N}(0, \sigma_i^2 + \sigma_{\rm int}^2)$.
- **Outliers** ($1 - f_{\rm in}$): contaminating sources with large, uninformative scatter $\sigma_{\rm out} \gg \sigma_{\rm int}$.

The mixture likelihood for star i is:

$$p(M_i \mid \boldsymbol{\theta}) = f_{\rm in}\, \mathcal{N}(M_i;\, a\,\delta_i + b,\, \sigma_i^2 + \sigma_{\rm int}^2) + (1 - f_{\rm in})\, \mathcal{N}(M_i;\, \mu_{\rm out},\, \sigma_{\rm out}^2). \quad (16)$$

Each star's posterior inlier probability $p_{{\rm in},i}$ is computed from the ratio of likelihood components. Stars with $p_{{\rm in},i} < 0.95$ were removed, reducing the sample from 973 to 874 stars. Figure 13 shows the PL relation before and after C1/C2 cuts, and Figure 14 shows the mixture-model inlier probabilities and the final cleaned sample.

The progressive improvement can also in quantified in scatter. The largest improvement occurs at the mixture model step where the RMS falls from 1.227 to 0.212 mag (a further 83% reduction), and the number of stars more than 1 mag from the relation drops from 49 to 0. The C1/C2 cuts alone provide a 51% RMS improvement, primarily by removing the long tails in the parallax error distribution. The resulting final calibration sample is shown in Figure 15.

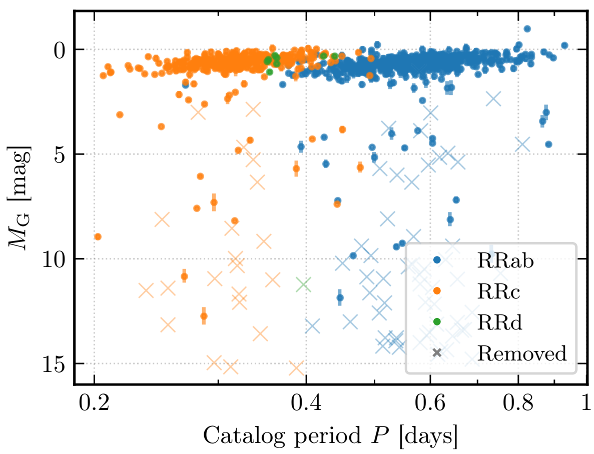

Fig. 13.— G-band period–absolute magnitude plot before (*left*, $N = 1031$) and after (*right*, $N = 973$) the Lindegren C1/C2 astrometric quality cuts. Points are colored by pulsation subclass. Error bars show combined photometric and parallax uncertainties.

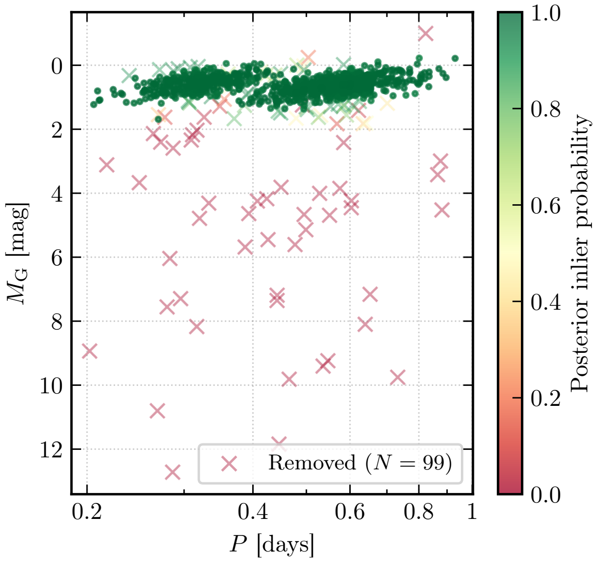

Fig. 14.— G-band period–absolute magnitude plot color-coded by posterior inlier probability. Circles show retained stars ($p_{\rm in} \geq 0.95$); crosses mark the flagged outliers ($p_{\rm in} < 0.95$) removed from the calibration sample.

Table 1: Calibration Sample Evolution Through Quality Cuts

Stage	N
Initial selection	1031
After C1/C2 cuts	973
After mixture model	874

6. Bayesian Period-Luminosity Fitting

6.1. Model Specification

We fit separate PL relations for RRab and RRc subclasses. The model for each subclass is:

$$M_G = a\,(\log P - \langle \log P \rangle_{\rm class}) + b + \epsilon_i, \qquad (17)$$

where $\langle \log P \rangle_{\rm class}$ is the mean $\log P$ for that subclass (RRab: $\langle \log P \rangle \approx -0.20$; RRc: $\langle \log P \rangle \approx -0.42$), and $\epsilon_i \sim \mathcal{N}(0, \sigma_i^2 + \sigma_{\rm scatter}^2 + a^2\sigma_{\log P,i}^2)$. Centering the $\log P$ variable makes the intercept b interpretable as the absolute magnitude at the class mean period and reduces parameter degeneracy.

We adopt weakly informative priors:

$$a \sim \mathcal{N}(0, 5^2), \qquad (18)$$

$$b \sim \mathcal{N}(\text{median}(M_G), 1^2), \qquad (19)$$

$$\sigma_{\rm scatter} \sim \text{HalfNormal}(0.5). \qquad (20)$$

The HalfNormal prior for $\sigma_{\rm scatter}$ is chosen because intrinsic scatter is a standard deviation and must be strictly non-negative; a full $\mathcal{N}(0, 0.5^2)$ prior would assign probability mass to unphysical negative values.

6.2. Three-Way MCMC Comparison

We implemented three MCMC methods for the RRab PL fit and compared their results:

Method 1 (M-H): Custom Metropolis-Hastings with diagonal Gaussian proposal. We ran 25,000 steps, discarding the first 5,000 as burn-in. The proposal widths are tuned by scanning a discrete grid; the achieved acceptance rate was 0.495.

Method 2 (NUTS + Potential): The gradient-based NUTS sampler in PyMC, with the log-likelihood

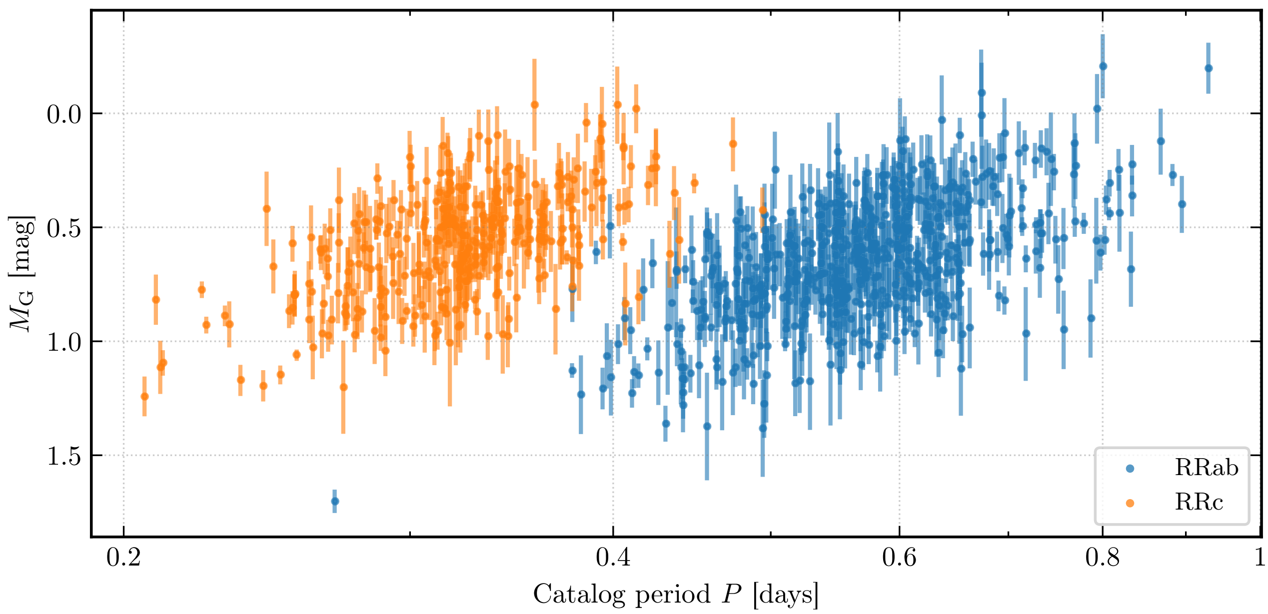

Fig. 15.— G-band period–absolute magnitude diagram for the final cleaned calibration sample ($N = 874$) after mixture contamination model rejection ($p_{\rm in} \geq 0.95$). Points are colored by pulsation subclass. Error bars show combined photometric and parallax uncertainties.

encoded as a `Potential` term. We ran 1,000 tuning steps and 2,000 draw steps.

Method 3 (Native PyMC): A fully probabilistic PyMC model with Gaussian likelihoods and HalfNormal priors, using the default NUTS sampler (1,000 tune, 2,000 draws).

Figures 16 and 17 overlay all three methods for direct comparison of the RRab and RRc posteriors respectively. All methods converge to consistent posterior means, but NUTS-based methods achieve much lower autocorrelation.

6.3. RRab Results

Using native PyMC NUTS on the 559 RRab stars in the cleaned calibration sample, we find:

$$a_{\rm RRab} = -2.2299 \pm 0.1261, \tag{21}$$
$$b_{\rm RRab} = +0.6523 \pm 0.0087, \tag{22}$$
$$\sigma_{\rm scatter,RRab} = 0.1752 \pm 0.0073. \tag{23}$$

Rather than overplotting individual posterior draws which can obscure the actual uncertainty structure by cluttering the figure, we display the posterior median relation together with 68% and 95% posterior predictive envelopes computed from the full post burn sample. This representation makes the shape and width of the uncertainty region immediately legible. The posterior predictive envelopes (shown in Figure 18) comfortably bracket the data with no systematic residual trends. The envelope width is approximately uniform across the full period range of the sample calibration stars the slope, and is tightly constrained ($\sigma_a \approx 0.13$). This is smaller than the intrinsic astrophysical scatter $\sigma_{\rm scatter} \approx 0.18$ mag, which dominates the envelope width throughout. The roughly constant band width therefore suggests that the uncertainty is set by possible stellar physics effects (metallicity spread, optical depth) rather than by parameter estimation uncertainty.

6.4. RRc Results

For the 315 RRc stars, the native PyMC NUTS fit yields:

$$a_{\rm RRc} = -2.2097^{+0.1727}_{-0.1794}, \tag{24}$$
$$b_{\rm RRc} = +0.5747 \pm 0.0115, \tag{25}$$
$$\sigma_{\rm scatter,RRc} = 0.1659 \pm 0.0093. \tag{26}$$

The slightly flatter slope for RRc compared to RRab is consistent with the different mass-luminosity relationship expected for first-overtone pulsators (Catelan & Smith 2015). Interestingly, the smaller intrinsic scatter ($\sigma_{\rm scatter,RRc} < \sigma_{\rm scatter,RRab}$) may also reflect the narrower instability strip occupied by first-overtone pulsators. The sampler comparison corner plot is shown in Figure 17.

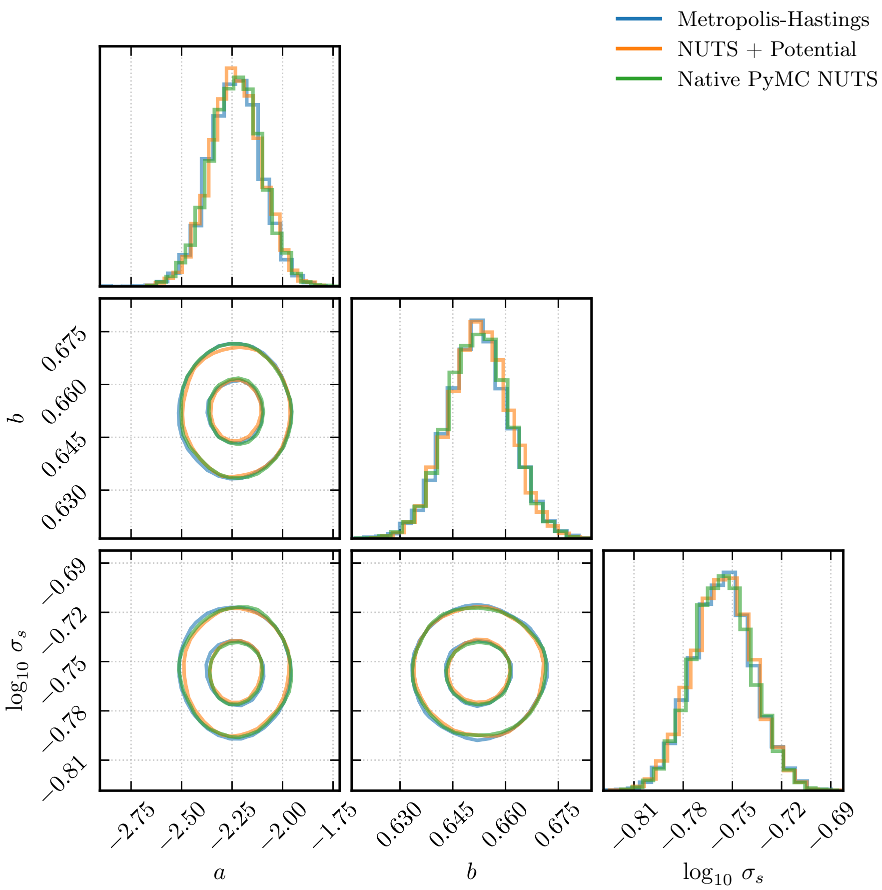

Fig. 16.— Overlay corner plot comparing all three MCMC methods for **RRab**: M-H (blue), NUTS+Potential (orange), native PyMC (green). All three converge to consistent posteriors, validating our implementation.

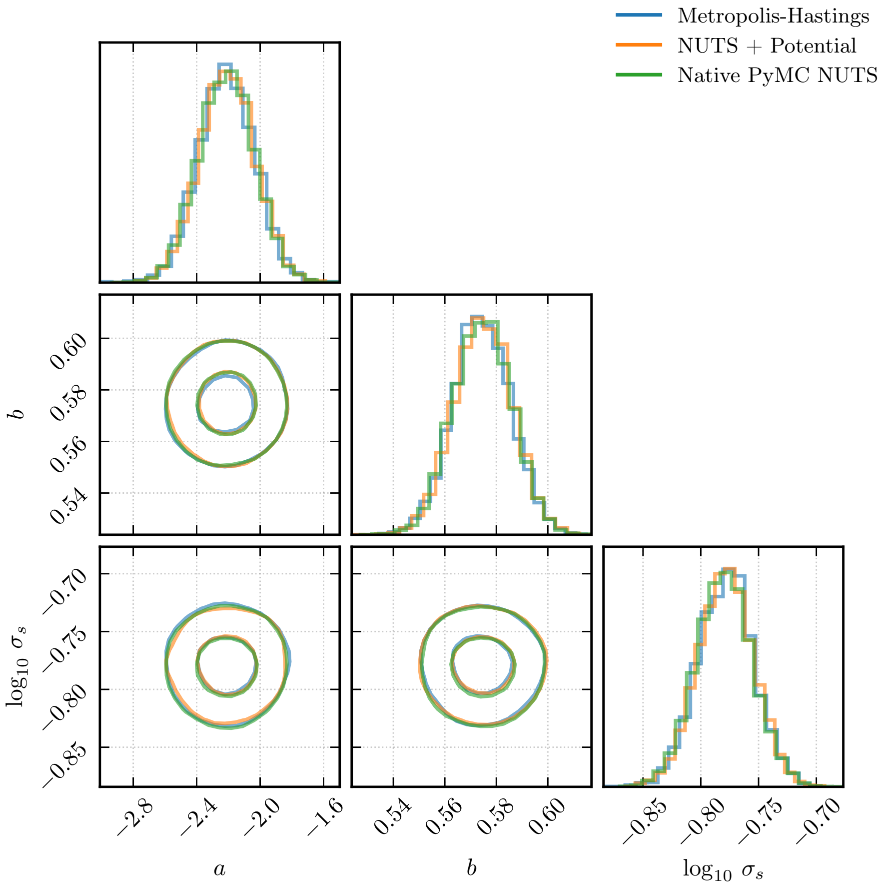

Fig. 17.— Overlay corner plot comparing all three MCMC methods for **RRc**: M-H (blue), NUTS+Potential (orange), native PyMC (green). The RRc posteriors are broader than RRab (fewer stars), but all three methods converge to consistent posteriors.

7. WISE W2 Cross-Match

7.1. Cross-Matching and Quality Cuts

WISE $W2$ photometry was retrieved for calibration sample stars via the Gaia DR3 `allwise_best_neighbour` table (Section 3). The cross-match proceeds in stages; first, the full Gaia DR3 × AllWISE join yields 3,615 rows; restricting to our clean 874-star calibration sample gives 830 matched stars with 526 RRab and 304 RRc. All 830 stars are used for the infrared analysis. The ∼5% cross-match loss (874 → 830) is consistent with the expected AllWISE completeness at these faint magnitudes. The sky distribution and period–absolute-magnitude cuts applied here mirror the G-band sample-selection procedure described in Section 5 (see Figures 12 and 13).

7.2. W2 Period-Luminosity Relations

We fit the $W2$ PL relation using the same Bayesian NUTS framework. The absolute $W2$ magnitude was computed from the apparent $W2$ magnitude and the Gaia parallax. Because $W2$ extinction is negligible ($A_{W2} \approx 0.02\, A_V$), no extinction correction was applied.

Figure 18 shows the posterior median and 68%/95% predictive envelopes over the RRab and RRc $W2$ data alongside the optical fits, and Table 2 summarizes the results. The slopes are significantly steeper than in G:

$$a_{W2,\mathrm{RRab}} = -2.9852 \pm 0.1147, \tag{27}$$
$$a_{W2,\mathrm{RRc}} = -3.2480 \pm 0.1561. \tag{28}$$

The intrinsic scatter is also smaller ($\sigma_{\mathrm{int}} \approx 0.14$–$0.15$ mag vs. ≈ 0.17–0.18 mag in G), reflecting both the reduced extinction sensitivity and tighter bolometric correction at $W2$ wavelengths.

7.3. Optical vs. Infrared Comparison

The infrared slope is steeper by $\Delta a_{\mathrm{RRab}} = -0.76 \pm 0.17$ and $\Delta a_{\mathrm{RRc}} = -1.04 \pm 0.24$. This steepening is a well-known property of RR Lyrae PL relations where at optical wavelengths, the luminosity-temperature-radius coupling partially cancels the period-luminosity correlation; in the infrared regime, the relationship becomes more purely between period and bolometric luminosity (Catelan & Smith 2015). Figure 18 shows both relations side by side for each subclass, making the slope difference directly visible.

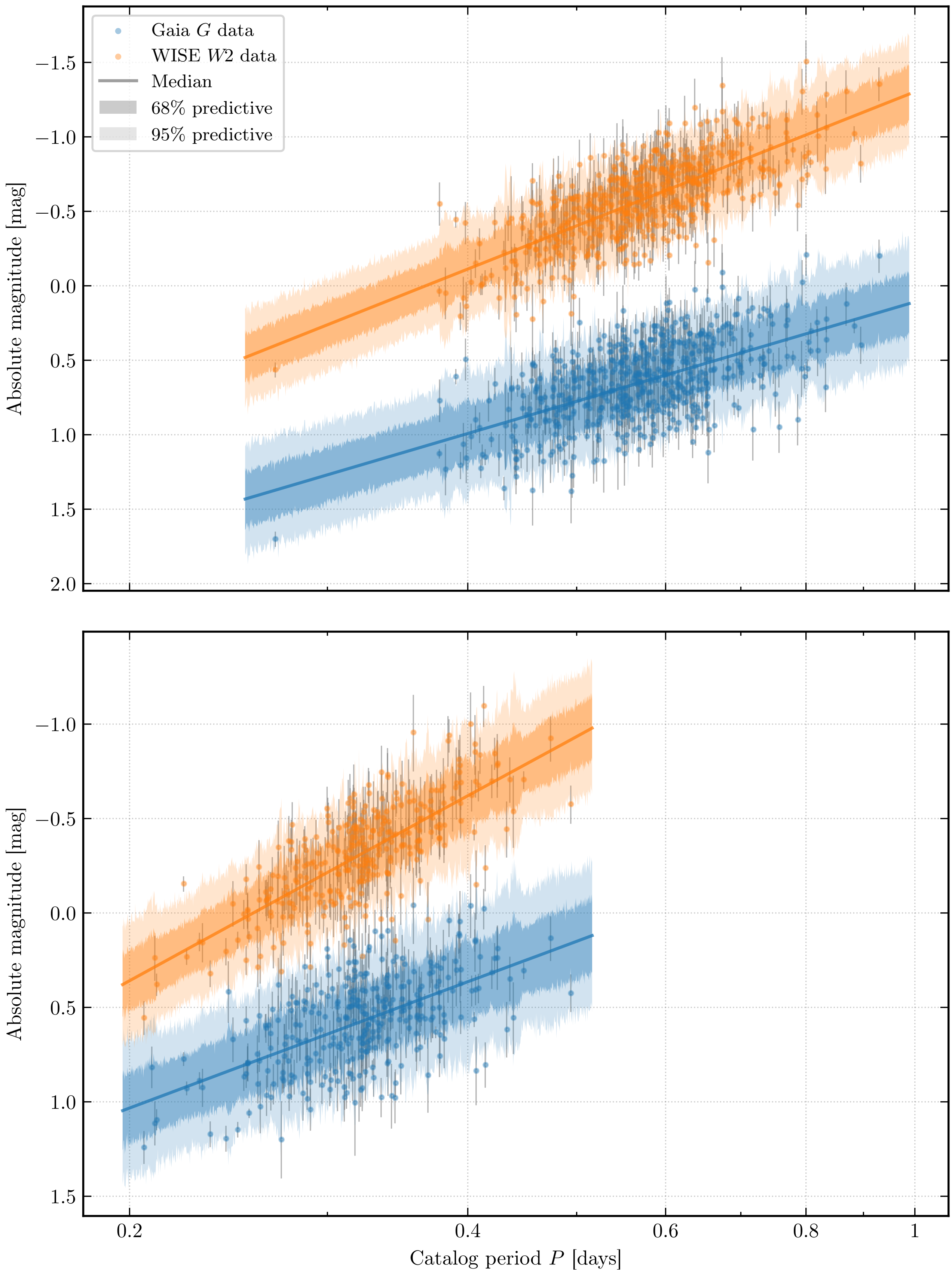

Fig. 18.— Side-by-side comparison of the Gaia G-band (blue) and WISE $W2$ (orange) period-luminosity relations for **RRab** (*top*) and **RRc** (*bottom*). Shaded bands show 68% posterior predictive envelopes. The steeper $W2$ slope ($\Delta a_{\mathrm{RRab}} = -0.76 \pm 0.17$, $\Delta a_{\mathrm{RRc}} = -1.04 \pm 0.24$) is visible as a more pronounced tilt in the orange fit lines relative to the blue. The tighter $W2$ scatter also reflects reduced extinction sensitivity at mid-infrared wavelengths.

Table 2: Period-Luminosity Relation Summary

Band	Type	a	b	σ_{int}
G (this work)	RRab (559)	-2.2299 ± 0.1261	0.6523 ± 0.0087	0.1752 ± 0.0073
G (this work)	RRc (315)	$-2.2097^{+0.1727}_{-0.1794}$	0.5747 ± 0.0115	0.1659 ± 0.0093
$W2$ (this work)	RRab (526)	-2.9852 ± 0.1147	$\cdots$	0.1467 ± 0.0074
$W2$ (this work)	RRc (304)	-3.2480 ± 0.1561	$\cdots$	0.1328 ± 0.0088
$W2$ (Klein et al. 2014)	RRab	-2.39 ± 0.20	$\cdots$	$\cdots$
$W2$ (Dambis et al. 2014)	RRab	-2.94 ± 0.15	$\cdots$	$\cdots$

8. Period-Color Relations

8.1. Model Specification

The color $G_{\text{BP}} - G_{\text{RP}}$ of a star depends on its temperature, which correlates with pulsation period. We model:

$$(G_{\text{BP}} - G_{\text{RP}}) = a_c \left(\log P - \langle \log P \rangle_{\text{class}}\right) + b_c + \epsilon_c, \quad (29)$$

where $\epsilon_c \sim \mathcal{N}(0, \sigma_c^2 + \sigma_{\text{int},c}^2)$, and σ_c is the measurement uncertainty on the color. We used only the 874-star cleaned calibration sample, for which residual extinction is assumed to be negligible (high Galactic latitude, low extinction region).

8.2. RRab Period-Color Relation

The NUTS fit for 559 RRab calibration stars gives:

$$a_{c,\text{RRab}} = +0.2483 \pm 0.0420, \quad (30)$$

$$b_{c,\text{RRab}} = +0.6487 \pm 0.0029, \quad (31)$$

$$\sigma_{c,\text{RRab}} = 0.0500 \pm 0.0026. \quad (32)$$

The positive slope means redder (cooler) RR Lyrae have longer periods, consistent with the width of the instability strip where lower-temperature stars pulsate at longer periods.

8.3. RRc Period-Color Relation

For 315 RRc stars:

$$a_{c,\text{RRc}} = +0.4783 \pm 0.0527, \quad (33)$$

$$b_{c,\text{RRc}} = +0.4452 \pm 0.0033, \quad (34)$$

$$\sigma_{c,\text{RRc}} = 0.0508 \pm 0.0026. \quad (35)$$

The steeper period-color slope for RRc compared to RRab (+0.48 vs. +0.25) suggests a stronger temperature-period coupling in first-overtone pulsation. The smaller intrinsic scatter ($\sigma_c \approx 0.05$ mag for both types) suggests that the period-color relation is tight, making it a reliable predictor of intrinsic color for individual stars.

Figure 19 shows the posterior predictive checks for

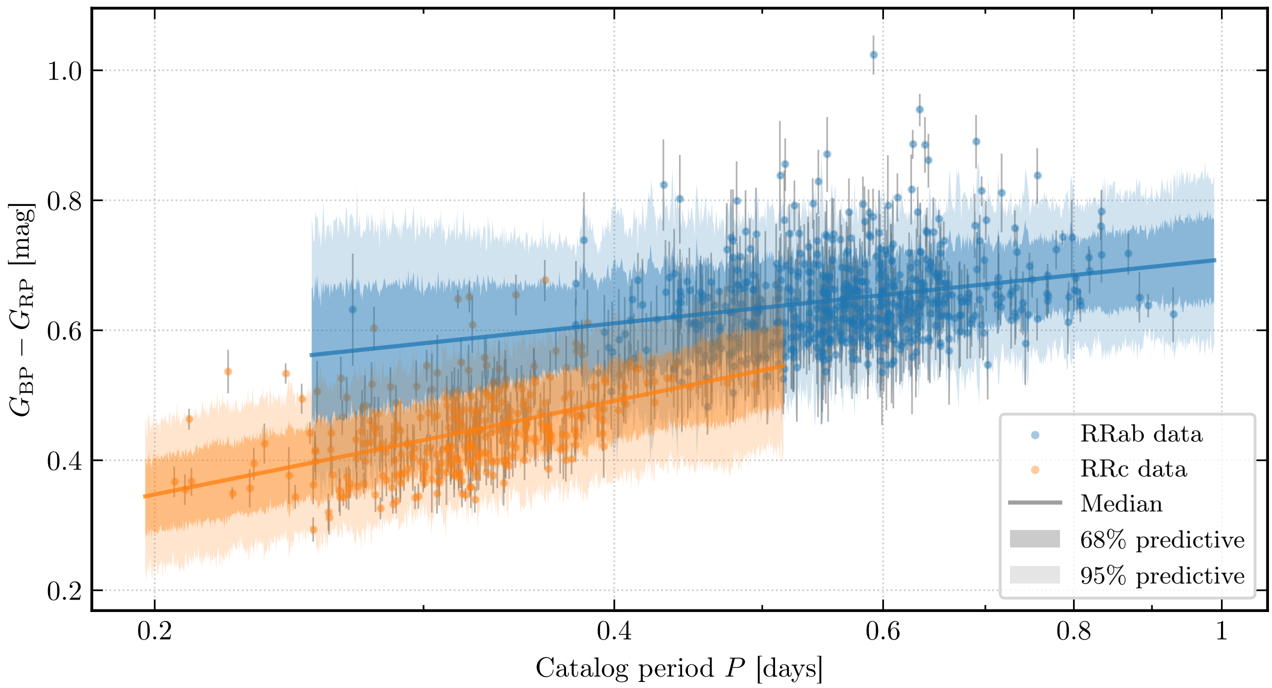

Fig. 19.— Joint period-color posterior predictive comparison for RRab (blue) and RRc (orange) in the same coordinate system. The RRc slope ($a_{c,\text{RRc}} = +0.48 \pm 0.05$) is roughly twice as steep as the RRab slope ($a_{c,\text{RRab}} = +0.25 \pm 0.04$), confirming stronger temperature-period coupling in first-overtone pulsators. Intercepts are evaluated at the class mean $\langle \log P \rangle$.

both subclasses, making the factor-of-two slope difference directly visible. The tight scatter ($\sigma_c \approx 0.05$ mag) suggests that RR Lyrae intrinsic colors are well-constrained by period alone.

9. Dust Mapping

9.1. Color Excess Calculation

For each of the 269,772 RR Lyrae stars in the full catalog, we computed the predicted intrinsic color using the period-color calibration from Section 8. The color excess is then:

$$E(G_{\rm BP} - G_{\rm RP}) = (G_{\rm BP} - G_{\rm RP})_{\rm obs} - a_c\,(\log P - \langle \log P\rangle_{\rm class}) - b_c, \quad (36)$$

and the G-band extinction:

$$A_G = R_G \cdot E(G_{\rm BP} - G_{\rm RP}), \quad R_G = 2.0. \quad (37)$$

The coefficient $R_G = 2.0$ is can be traced to be adopted following Wang & Chen (2019) and is consistent with the Cardelli et al. (1989) extinction law as updated by O'Donnell (1994) integrated over the Gaia G bandpass. This approach traces the dust integrated along the line of sight out to the distance of each star, providing inherently three-dimensional extinction information.

9.2. Comparison to Gaia DR3 `g_absorption`

As a sanity check, we compared our empirical A_G estimates to the Gaia DR3 `g_absorption` field, which uses the Gaia photometric solution to estimate extinction. Figure 20 shows this comparison where we find that our median offset is 0.146 mag lower than `g_absorption`.

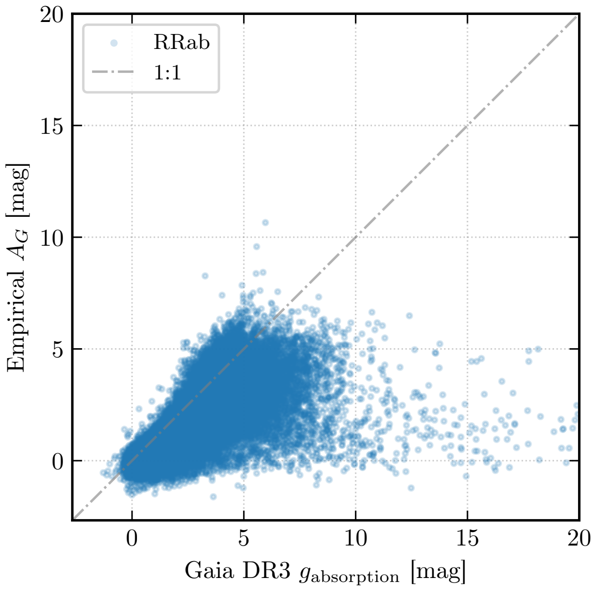

Fig. 20.— Empirical A_G (this work) vs. Gaia DR3 `g_absorption` for 142,867 RRab stars with finite Gaia extinction estimates. The dashed line shows the 1:1 relation.

Compared to the Gaia DR3 `g_absorption` which is computed from a global photometric model that may be miscalibrated at low extinction levels, our empirical approach is anchored to the observed period-color relation. The systematic offset could also reflect a unaccounted for calibration zero-point difference in R_G or the period-color

9.3. Quality Cuts for the Sky Map

To ensure a clean reddening map, we applied the following quality cuts to the full 269,772-star catalog:

1. `phot_bp_mean_flux_over_error` > 5 (good BP photometry): removes 35,169 stars (13.0%)
2. `phot_rp_mean_flux_over_error` > 5 (good RP photometry): removes 1,515 stars (0.6%)
3. BP/RP photometric excess within the magnitude-dependent envelope (similar to C2 cut): removes 55,304 stars (20.5%), primarily heavily blended sources concentrated toward the Galactic plane with bad photometric solutions
4. Uncertainty on color excess $\sigma_E \leq 0.15$ mag: removes 8,000 stars (3.0%)
5. $0 \leq E(G_{\rm BP} - G_{\rm RP}) \leq 10$ mag (exclude unphysical negative reddening): removes 38,571 stars (14.3%), primarily low-latitude stars with negative empirical reddening. The upper bound sanity cap of 10 mag affects no stars passing the preceding quality cuts

After these cuts, 131,213 stars (48.6% of the initial catalog) remained for the sky map. The removed stars are not random but concentrated toward the Galactic plane and other crowded or high-extinction regions.

9.4. All-Sky Reddening Map

Figure 21 shows the all-sky reddening map in Aitoff projection and figure 22 shows the SFD reddening map.

From the map, we can identify that (i) the highest reddening is concentrated along the Galactic plane, particularly toward the Galactic center; (ii) even at high latitudes, there are patchy low-level reddening features from local dust clouds; and (iii) the RR Lyrae distribution is not uniform across the sky, with overdensities corresponding to nearby dwarf galaxies with particularly the Magellanic Clouds appear as overdensities and underrepresentation in the inner disk and bulge due to high extinction.

This non-uniform sampling is an inherent limitation of using RR Lyrae as dust tracers since they trace the dust only where RR Lyrae stars exist, which preferentially samples old stellar populations in the halo and thick disk. Completeness is further reduced toward the Galactic plane by extinction itself, creating a selection bias where the most heavily extincted regions are least well-sampled.

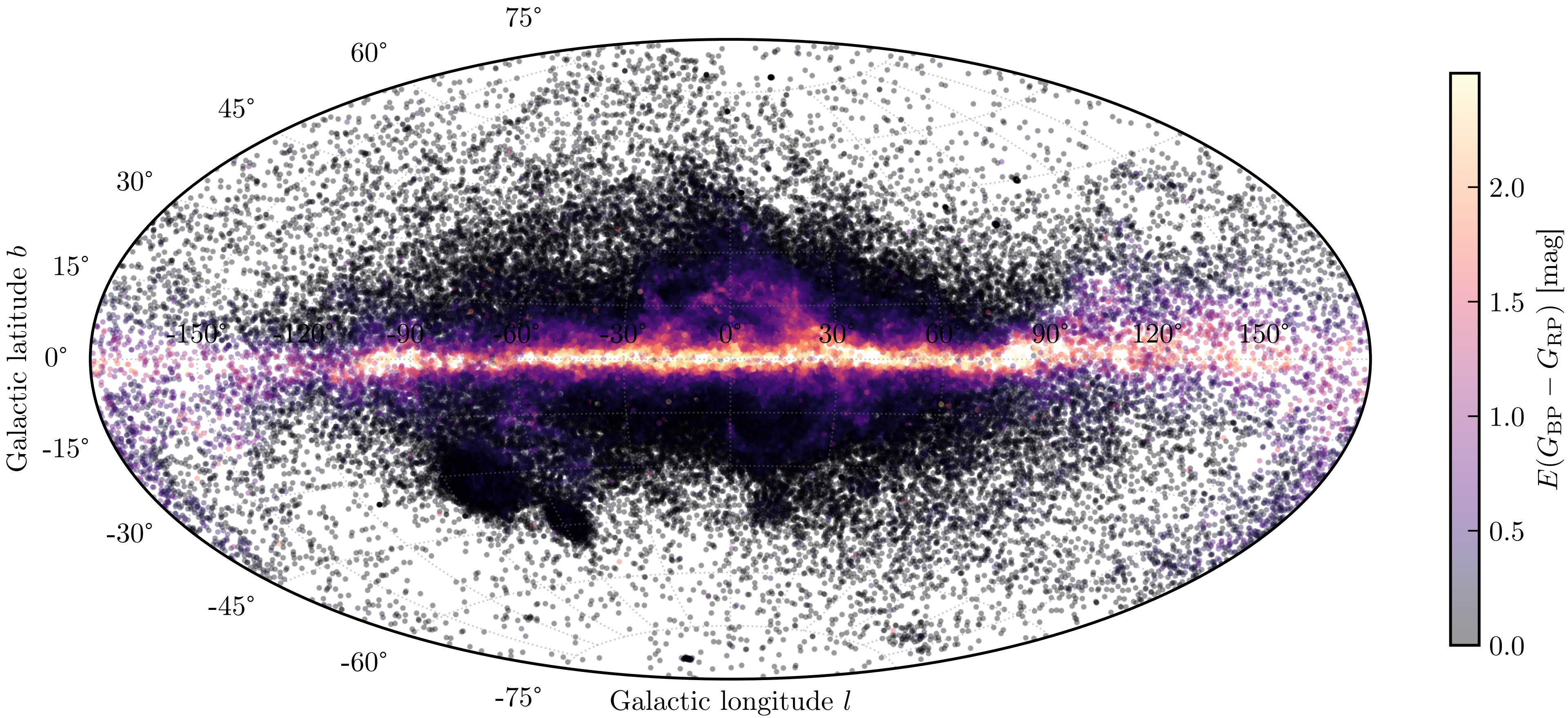

Fig. 21.— All-sky reddening map from 131,213 RR Lyrae stars (88,079 RRab + 43,134 RRc) in Aitoff (equal-area) projection. Color indicates $E(G_{\rm BP} - G_{\rm RP})$ in magnitudes. The Galactic plane shows high reddening; the Galactic center region (right) has the highest extinction. High-latitude regions show low, patchy reddening from nearby dust clouds; the Large and Small Magellanic Clouds appear clearly as overdensities. The non-uniform sky coverage reflects the non-uniform distribution of RR Lyrae stars.

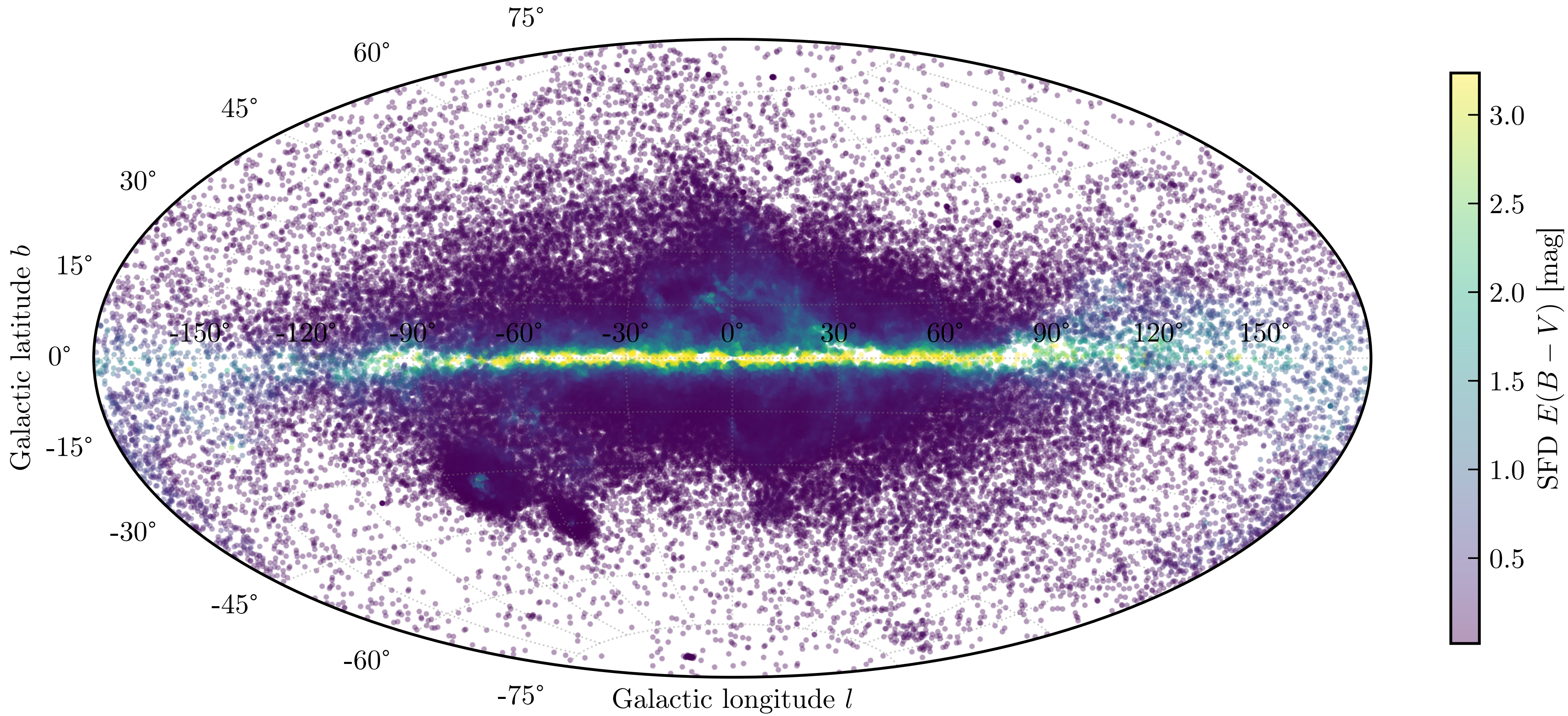

Fig. 22.— SFD $E(B-V)$ (Schlegel et al. 1998) sampled at the sky positions of the 131,213 cleaned RR Lyrae stars in Aitoff (equal-area) projection. The map traces the total dust column integrated to infinity along each sightline; compare with Figure 21, which shows only the foreground dust out to each star's heliocentric distance. The Galactic plane and center show markedly higher $E(B-V)$ in SFD than in the empirical RR Lyrae map which provides direct evidence of the 3D vs. 2D effect explored quantitatively in Section 9.6.

9.5. Comparison to the SFD Map

We compared our RR Lyrae-based reddening to the SFD dust map (Schlegel et al. 1998) by evaluating the SFD $E(B-V)$ at the sky position of each RR Lyrae star.

We quantify agreement between the two maps using the Pearson R^2 coefficient, which measures the fraction of variance in one reddening estimate explained by a linear fit to the other. Since, R^2 is invariant to rescaling of either axis, the Gaia-band conversion factor $R_{BP-RP} \approx 1.3$ between $E(G_{\rm BP}-G_{\rm RP})$ and $E(B-V)$ should not affect the result.

Table 3: Pearson R^2 with SFD by Galactic Latitude (10° bins; $|b| > 60°$ split into finer bins)

Latitude range	$N_{\rm stars}$	R^2
$\|b\| < 10°$	54,515	0.3467
$10° < \|b\| < 20°$	33,393	0.8178
$20° < \|b\| < 30°$	17,385	0.7641
$30° < \|b\| < 40°$	18,168	0.0540
$40° < \|b\| < 50°$	5,066	0.1645
$50° < \|b\| < 60°$	1,440	0.0334
$60° < \|b\| < 70°$	682	0.0263
$70° < \|b\| < 80°$	413	0.0113
$\|b\| > 80°$	151	0.0000
Overall	131,213	0.457

The overall $R^2 = 0.457$ suggests the period-color approach for dust tracing is effective. Interestingly, R^2 peaks in the intermediate-latitude bins ($R^2 = 0.8178$ at 10°–20° and 0.7641 at 20°–30°) rather than at the lowest latitudes ($R^2 = 0.3467$ at $|b| < 10°$), and drops sharply at high latitudes. The finer breakdown at high latitudes (Table 3) shows that R^2 falls below 0.03 between 60° and 80° and shows no meaningful signal at the poles (151 stars at $|b| > 80°$). This latitude dependence is physically expected: near the plane, the dust column is large (up to $E(B-V) \sim 5$ mag), providing a large dynamic range that allows both methods to correlate well. At high latitudes, the dust column is small ($E(B-V) \lesssim 0.05$ mag) and both measurements are noise-dominated.

9.6. SFD Regime Decomposition Comparison

The latitude-binned R^2 table characterizes the global agreement, but the physical behavior is more sharply revealed by splitting on the magnitude of SFD $E(B-V)$ rather than by latitudes. We define two regimes:

1. **Similar-scale** ($E(B-V)_{\rm SFD} \leq 2$ mag, $N =$ 128,154 stars): SFD and the empirical RR Lyrae color excess are in the same range.
2. **Large-SFD** ($E(B-V)_{\rm SFD} > 10$ mag, $N =$

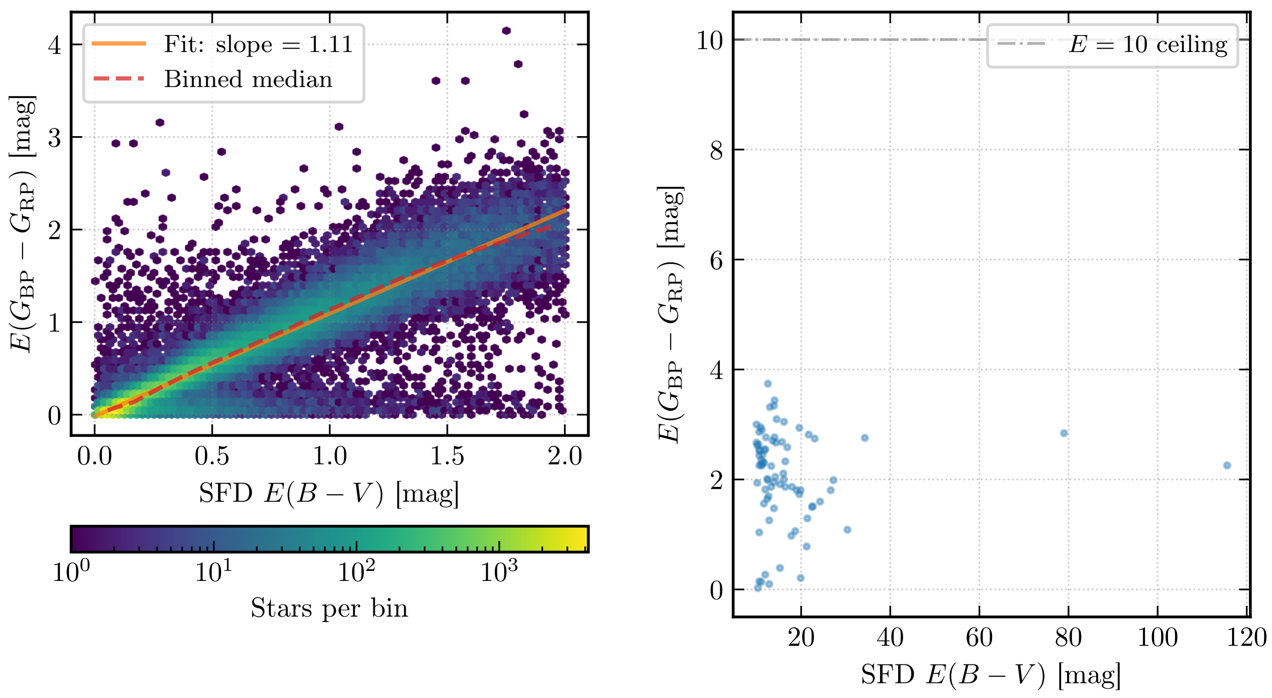

Fig. 23.— Regime decomposition of the SFD vs. empirical comparison. *Left:* Similar-scale regime ($E(B-V)_{\rm SFD} \leq 2$ mag, $N = 128{,}154$ stars). The density hexbin shows a tight linear trend; the dashed orange line is a linear fit with slope 1.11, below the expected Gaia-band conversion factor $R_{BP-RP} \approx 1.3$. *Right:* Large-SFD regime ($E(B-V)_{\rm SFD} > 10$ mag, $N = 77$ stars). The empirical $E(G_{\rm BP} - G_{\rm RP})$ saturates far below the SFD column despite the physical ceiling at 10.0 mag being well above the observed values. The decoupling illustrates the 3D vs. 2D distinction between a distance-limited stellar measurement and a total-column dust map.

77 stars): these are the most heavily obscured Galactic plane and inner-Galaxy sightlines.

Figure 23 shows a two-panel summary. In the similar-scale panel (left), a hexbin density plot combined with a linear fit reveals a slope of 1.11, somewhat below the expected Gaia-band reddening conversion factor $R_{BP-RP} \equiv E(G_{\rm BP} - G_{\rm RP})/E(B - V) \approx 1.3$ for a standard $R_V = 3.1$ extinction law (Fitzpatrick 1999; Wang & Chen 2019). The deviation from unity is consistent with the 3D vs. 2D effect where in this moderate-reddening subsample, RR Lyrae stars at finite distances measure only a foreground fraction of the SFD column. Furthermore, the Pearson R^2 within this subsample is 0.877, and the binned median trend follows the linear fit closely, confirming that the empirical period-color calibration recovers a consistent reddening signal relative to SFD across the entire moderate-extinction range.

In the large-SFD panel (right), the empirical values cluster well below the physical ceiling while SFD spans the full range above 10 mag despite significantly higher SFD magnitudes.

A possible explanation for this apparent saturation is two-fold. Since RR Lyrae stars lie at finite heliocentric distances (typically 1–30 kpc). SFD integrates dust to infinity. In the deeply obscured Galactic plane, the bulk of the dust column lies beyond the star, so the stellar measurement captures only a foreground fraction of the SFD column. Additionally, stars toward heavily obscured sightlines tend to have poor BP/RP signal-to-noise and anomalous BP/RP excess, causing them to fail the quality filters described in Section 9. The surviving sample is already biased toward less-reddened, foreground stars. Together, this result contrasts the regime where a 2D total-column map (SFD) and a 3D distance-limited stellar measurement (this work) are expected to diverge. A full-3D dust map such as Bayestar (Green et al. 2019) is required to resolve this degeneracy.

10. Results and Discussion

10.1. Period-Luminosity Relations

Table 2 summarizes our PL relations. The G-band slopes for RRab and RRc are consistent within 2σ, though RRab are slightly steeper. In optical bands, RRab and RRc occupy different parts of the instability strip and may follow subtly different PL relations due to their different mass-luminosity relationships.

The intrinsic scatter $\sigma_{\rm scatter} \approx 0.17$–$0.18$ mag in G could be dominated by the unmodelled metallicity spread, Z_λ[Fe/H].

Our $W2$ RRab slope (-2.99 ± 0.11) lies $\sim 2.6\sigma$ from the Klein et al. (2014) value ($a = -2.39 \pm 0.20$), possibly reflecting different sample selections or parallax zero-points. It agrees within 1σ with the pure PL slope reported by Dambis et al. (2014) ($a = -2.94 \pm 0.15$). The small differences may reflect (i) different metallicity distributions between samples, (ii) different parallax zero-point conventions (Gaia EDR3 vs. HST or other calibrators), or (iii) different reference period choices. For RRc stars, Klein et al. (2014) report $a_{W2,\rm RRc} = -1.70 \pm 0.62$, which is substantially shallower than our value of -3.25 ± 0.16. The large uncertainty on the Klein et al. (2014) RRc slope (± 0.62) reflects their much smaller sample size. Compared to Klein et al. (2014), our RRc slope deviates at the $\sim 2.5\sigma$ level.

Our G-band slopes are somewhat steeper than optical V-band literature values (typically $a_V \approx -1.4$ to -1.8; Beaton et al. 2018), as expected since G is redder than V and should have a steeper period-luminosity slope. Beaton et al. (2018) present V-band RR Lyrae period-luminosity relations from the extragalactic Carnegie–Chicago Hubble Program with slopes in the range $a_V \approx -1.4$ to -1.8 and intrinsic scatter of $\sim$0.14–0.20 mag, which they use to calibrate the extragalactic distance scale. The G-band is broader and extends redward of V, so the Gaia G-band PL slope is expected to be steeper, as we observe ($a_G \approx -2.2$ vs. $a_V \approx -1.5$). The larger intrinsic scatter we find in G ($\sigma_{\rm scatter} \approx 0.17$ mag) compared to their V-band values is also consistent with the increased sensitivity of optical bands to metallicity spread.

Literature values are listed in Table 2 for direct numerical comparison but are *not* overlaid on the posterior predictive check figures (Figures 18 and 19). Those figures display our model evaluated over our own calibration sample; overlaying a literature relation calibrated on a different dataset, with different parallax zero-points, and metallicity distributions, would conflate the intrinsic scatter of our data with the systematic offset between datasets and produce a misleading visual comparison.

10.2. Wavelength Dependence

The $W2$ slopes are steeper by $\Delta a \approx -0.76$ (RRab) and ≈ -1.04 (RRc) compared to G. This confirms the well-established wavelength dependence of RR Lyrae PL relations. The reduced scatter in $W2$ ($\sigma_{\rm scatter} \approx 0.14$ mag) has two contributions: (i) the metallicity effect on luminosity is smaller in the infrared (Catelan & Smith 2015), and (ii) the $W2$ observations are unaffected by residual extinction in our sample, which at $|b| > 30^\circ$ amounts to $A_G \approx 0.0$–0.2 mag.

10.3. Dust Map Quality and Limitations

Our 131,213-star reddening map is a large RR Lyrae-based dust maps compiled from Gaia data. Primarily, it is limited by sampling inhomogeity and distance-limited coverage of RR Lyrae stars observable. RR Lyrae are old ($>$ 10 Gyr) metal-poor stars that trace the Galactic halo and bulge. Their sky distribution is thus non-uniform in a physically convenient sense since they concentrate toward the Galactic center (bulge overdensity) and are distributed throughout the halo, but are severely underrepresented in the Galactic plane ($|b| \lesssim 10°$) because the dense dust column makes RR Lyrae in the plane observationally inaccessible in the optical G band. The outer disk and spiral arms, which host young stellar populations but very few old metal-poor halo stars, and thus are also poorly sampled. This contrasts with dust, which follows the young stellar population and interstellar gas more closely.

Additionally, most stars in our sample from Gaia with good astrometric data are within ∼10 kpc. The SFD map, derived from far-infrared emission, includes all Galactic dust to infinity. Our map will therefore underestimate extinction toward the Galactic center and beyond the outer disk.

Lastly, we also note that the maps look patchy with missing gaps of data. With 131,213 stars over the full sky, the mean angular separation between stars is $\sim 0.3°$. The effective angular resolution of the map is thus limited to ∼0.3–0.5 degrees, comparable to the SFD resolution of $\sim 6'$ but with much noisier individual measurements.

Despite these limitations, the strong R^2 with SFD demonstrates that RR Lyrae are effective dust tracers.

11. Conclusions

We have constructed an empirical all-sky dust reddening map from Gaia DR3 RR Lyrae stars using a pipeline that proceeds from raw light curve data through period recovery, Fourier modeling, calibration, and Bayesian inference to a 131,213-star sky map.

Period recovery from Lomb-Scargle periodograms agrees with Gaia catalog values to within 1% for 89 of 100 test stars. All 11 outlying cases arise because $P_{\rm LS}$ latches onto the first-overtone period `p1o` rather than the fundamental `pf`, identifying these stars as RRd double-mode pulsators. Blazhko-effect modulations, though visible in some phase-folded light curves, carry negligible LS power and do not affect period recovery.

Mean magnitudes are estimated by integrating cross-validated Fourier series fits (optimal order $K \approx 5$–7) in flux space. This reduces the RMS from 0.0282 to 0.0139 mag relative to naïve epoch-averaging, owing to the non-linear magnitude–flux relation and the asymmetric sawtooth shape of RRab light curves.

The calibration sample was assembled from 1031 nearby ($d < 4$ kpc), high-latitude ($|b| > 30°$) stars and cleaned through Lindegren C1/C2 astrometric quality cuts and a two-component mixture contamination model, arriving at 874 stars (559 RRab + 315 RRc). G-band period-luminosity relations fit by Bayesian NUTS sampling give $a_{\rm RRab} = -2.23 \pm 0.13$, $b_{\rm RRab} = 0.652 \pm 0.009$ and $a_{\rm RRc} = -2.21 \pm 0.18$, $b_{\rm RRc} = 0.575 \pm 0.011$, with three independent MCMC approaches (Metropolis-Hastings, NUTS with a custom potential, native PyMC) yielding mutually consistent posteriors.

Extending to WISE $W2$ photometry, we find slopes steeper by $\Delta a \approx -0.76$ (RRab) and ≈ -1.04 (RRc) relative to G, with reduced intrinsic scatter ($\sigma_{\rm scatter} \approx 0.14$ mag), consistent with the well-established wavelength dependence of RR Lyrae period-luminosity relations and the near-zero $W2$ extinction coefficient. The RRab $W2$ slope (-2.99 ± 0.11) agrees within 1σ with Dambis et al. (2014) ($a = -2.94 \pm 0.15$). Period-color relations in $G_{\rm BP} - G_{\rm RP}$ are tight ($\sigma_c \approx 0.05$ mag) with slopes $a_{c,\rm RRab} = +0.25 \pm 0.04$ and $a_{c,\rm RRc} = +0.48 \pm 0.05$.

Applying the period-color calibration to the full 269,772-star catalog and retaining 131,213 stars (88,079 RRab + 43,134 RRc) after quality filtering yields an all-sky reddening map with Pearson $R^2 = 0.457$ against the SFD map overall and $R^2 = 0.877$ in the similar-scale regime ($E(B-V)_{\rm SFD} \leq 2$ mag, $N = 128{,}154$). A regime decomposition reveals a linear fit slope of 1.11 in that subsample, below the expected $R_{BP-RP} \approx 1.3$, consistent with RR Lyrae at finite distances measuring only a foreground fraction of the total SFD column. In the large-SFD regime ($E(B-V)_{\rm SFD} > 10$ mag, $N = 77$), empirical reddening values saturate well below the SFD column, directly illustrating the 3D vs. 2D distinction inherent to stellar dust tracers.

Future work can incorporate spectroscopic metallicities (e.g., from APOGEE cross-matches) to add a metallicity term to the PL relation, reducing intrinsic scatter. The upcoming Gaia DR4 and DR5 releases will increase the catalog size, improving map resolution and depth. Together with a larger dataset, a joint hierarchical model combining parallax, period, photometry, and metallicity in a single inference step would further improve both the distance estimates and the dust map precision.

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A. Reproducibility Notes

All code, notebooks, cached analysis products, and report sources used for this lab are version-controlled at `https://github.com/raw-asparagus/ay-128`.